University of Salford MANCHESTER

BEng (Hons) Control and Automation

Automation and Robotics

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Lecture Content:

Second Order Systems

and

PID controllers

Introduction

We have already discussed the affect of location of the pole on the transient response of 1st order systems.

Compared to the simplicity of a first-order system, a second-order system exhibits a wide range of responses that must be analyzed and described.

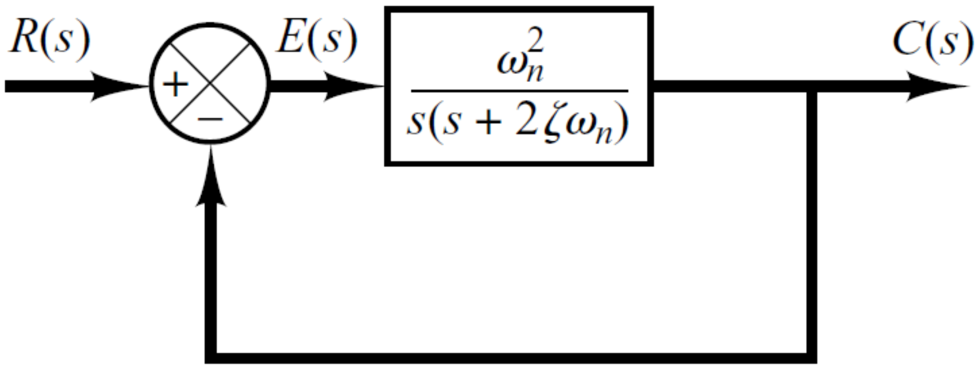
Varying a first-order system's parameters (T, K) simply changes the speed and offset of the response

Whereas, changes in the parameters of a second-order system can change the *form* of the response.

A second-order system can display characteristics much like a first-order system or, depending on component values, display damped or *pure oscillations* for its *transient response*.

Introduction

A general second-order system (without zeros) is characterized by the following transfer function.



$$G(s) = \frac{\omega_n^2}{s(s+2\zeta\omega_n)}$$

Open-Loop Transfer Function

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Closed-Loop Transfer Function

Introduction

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

ζ $\longrightarrow$ damping ratio of the second order system, which is a measure of the degree of resistance to change in the system output.

ω_n $\longrightarrow$ un-damped natural frequency of the second order system, which is the frequency of oscillation of the system without damping.

Example

- Determine the un-damped natural frequency and damping ratio of the following second order system.

$$\frac{C(s)}{R(s)} = \frac{4}{s^2 + 2s + 4}$$

- Compare the numerator and denominator of the given transfer function with the general 2nd order transfer function.

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_n^2 = 4 \Rightarrow \omega_n = 2\ rad/\text{sec}$$

$$\cancel{s^2} + 2\zeta\omega_n s + \cancel{\omega_n^2} = \cancel{s^2} + 2s + \cancel{4}$$

$$\Rightarrow 2\zeta\omega_n s = 2s$$

$$\Rightarrow \zeta\omega_n = 1$$

$$\Rightarrow \zeta = 0.5$$

Introduction

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- The closed-loop poles of the system are

$$-\omega_n\zeta + \omega_n\sqrt{\zeta^2 - 1}$$

$$-\omega_n\zeta - \omega_n\sqrt{\zeta^2 - 1}$$

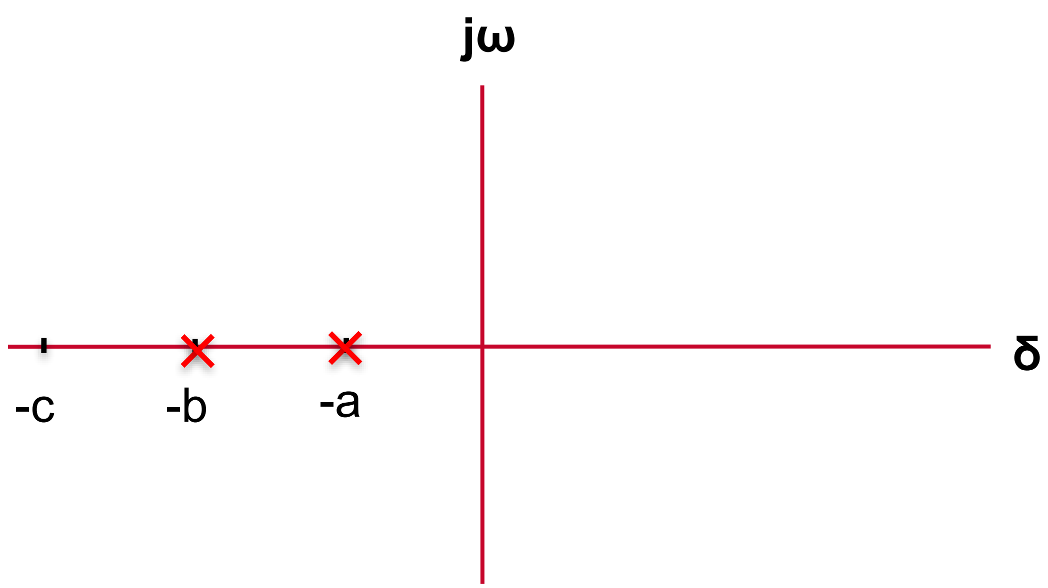
Introduction

$$-\omega_n\zeta + \omega_n\sqrt{\zeta^2 - 1}$$

$$-\omega_n\zeta - \omega_n\sqrt{\zeta^2 - 1}$$

- Depending upon the value of ζ, a second-order system can be set into one of the four categories:

1. *Overdamped* - when the system has two real distinct poles (ζ >1).



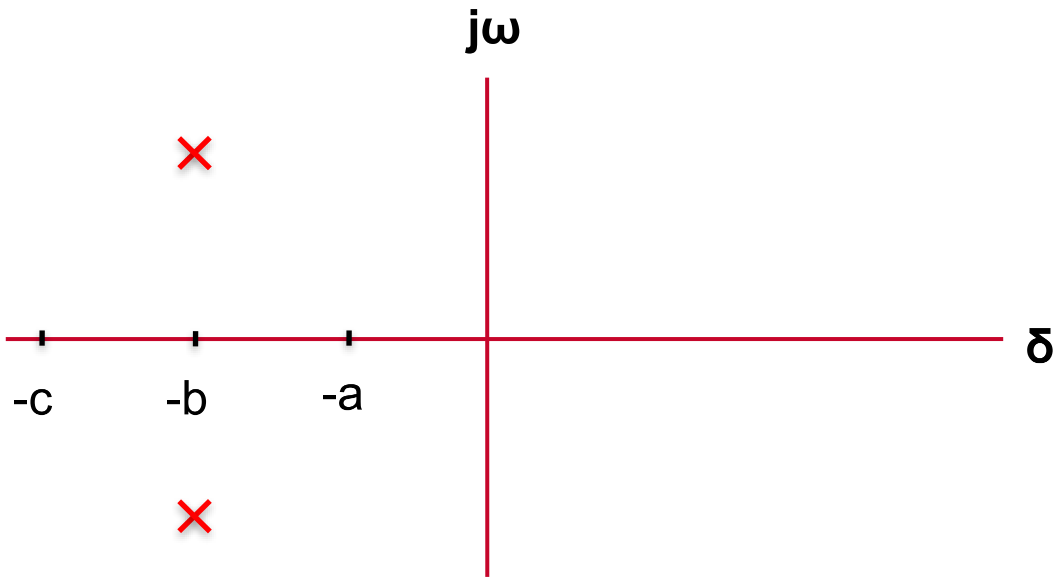
Introduction

$$-\omega_n\zeta + \omega_n\sqrt{\zeta^2 - 1}$$

$$-\omega_n\zeta - \omega_n\sqrt{\zeta^2 - 1}$$

- Depending upon the value of ζ, a second-order system can be set into one of the four categories:

2. *Underdamped* - when the system has two complex conjugate poles ($0 < \zeta < 1$)

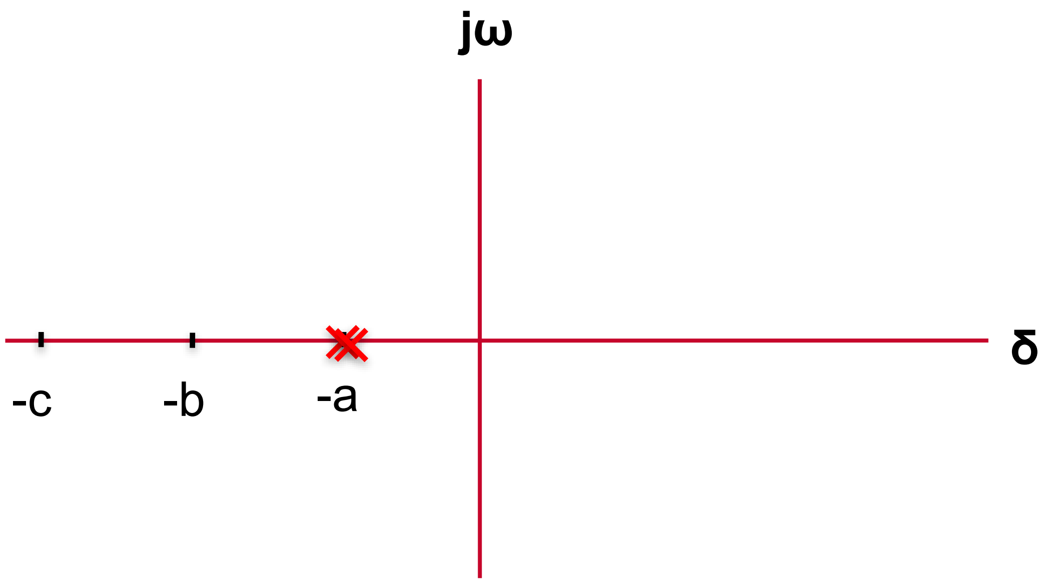


Introduction

$$-\omega_n\zeta + \omega_n\sqrt{\zeta^2 - 1}$$
$$-\omega_n\zeta - \omega_n\sqrt{\zeta^2 - 1}$$

- Depending upon the value of ζ, a second-order system can be set into one of the four categories:

3. *Critically damped* - when the system has two real but equal poles (ζ = 1).



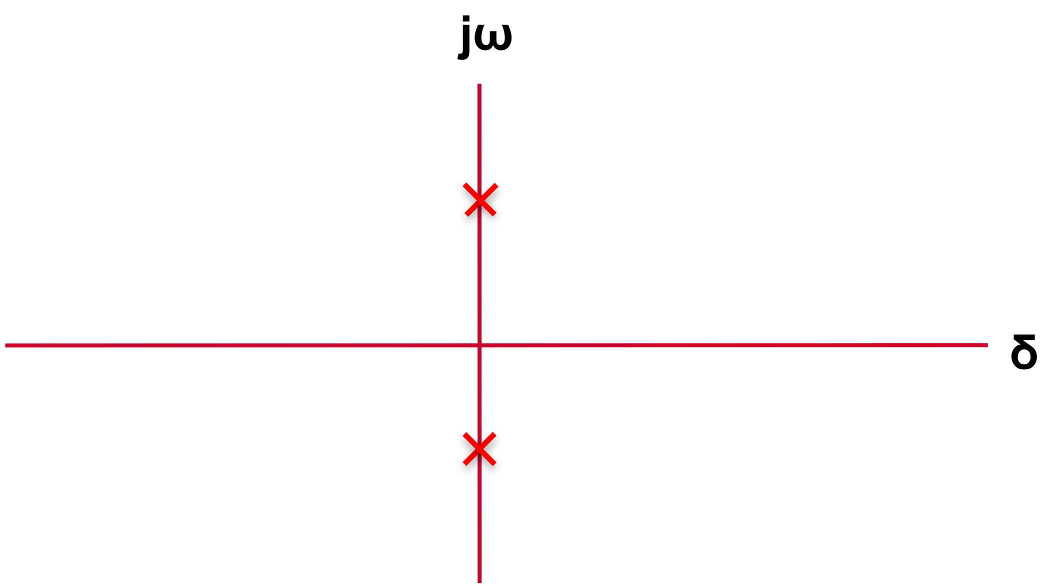
Introduction

$$-\omega_n\zeta + \omega_n\sqrt{\zeta^2 - 1}$$

$$-\omega_n\zeta - \omega_n\sqrt{\zeta^2 - 1}$$

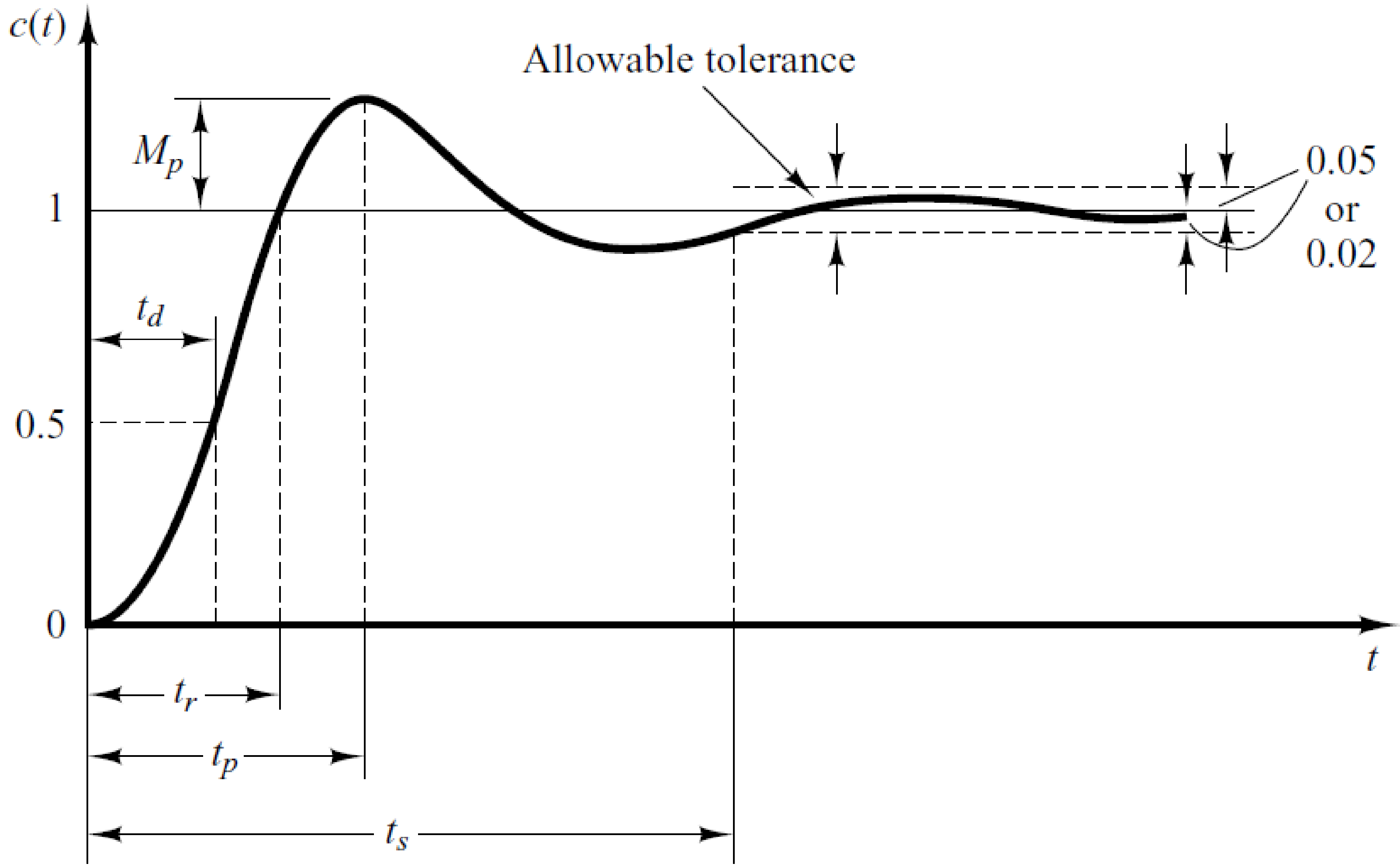
- Depending upon the value of ζ, a second-order system can be set into one of the four categories:

4. *Undamped* - when the poles have no real component.

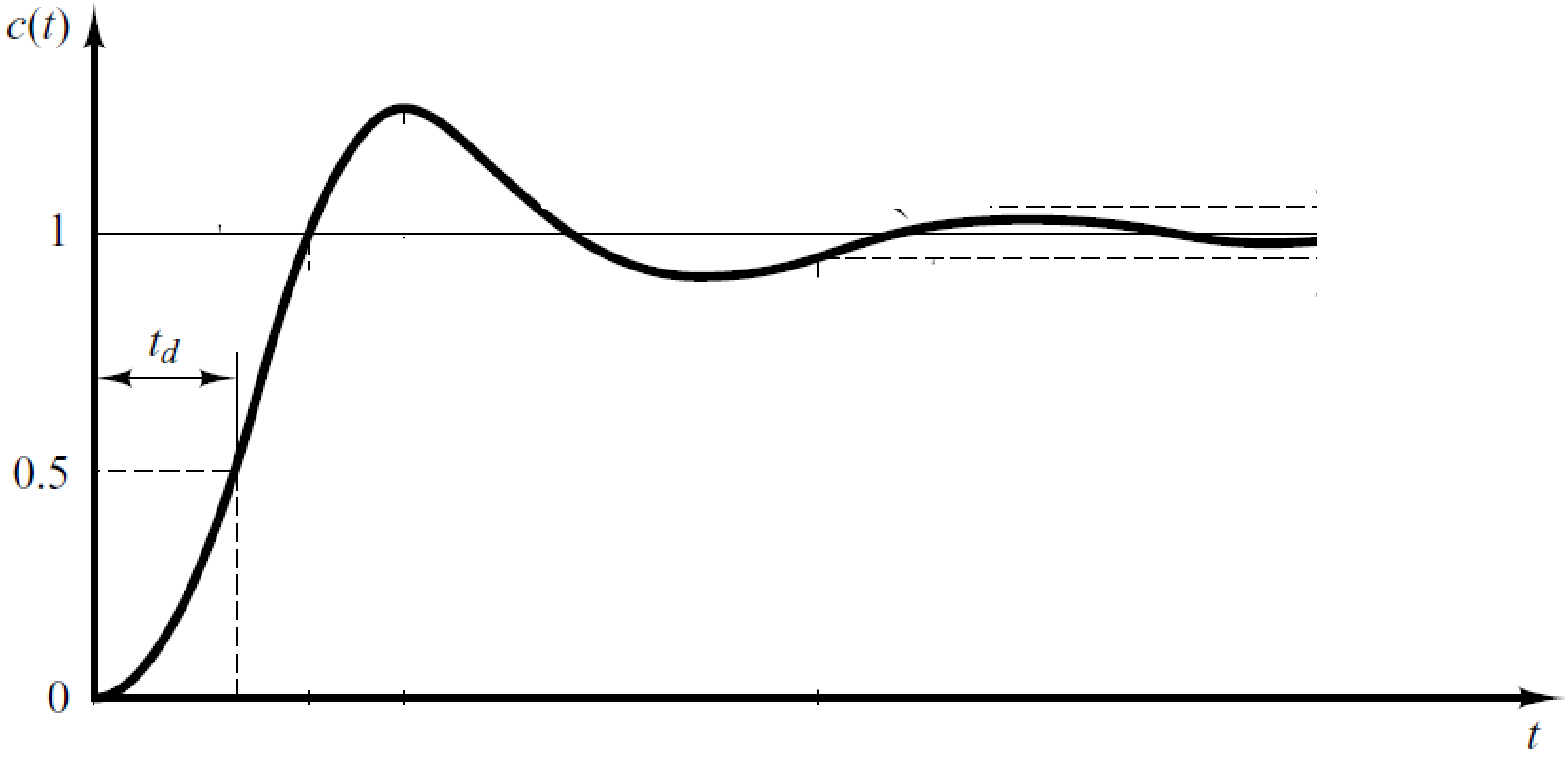


Time-Domain Specification

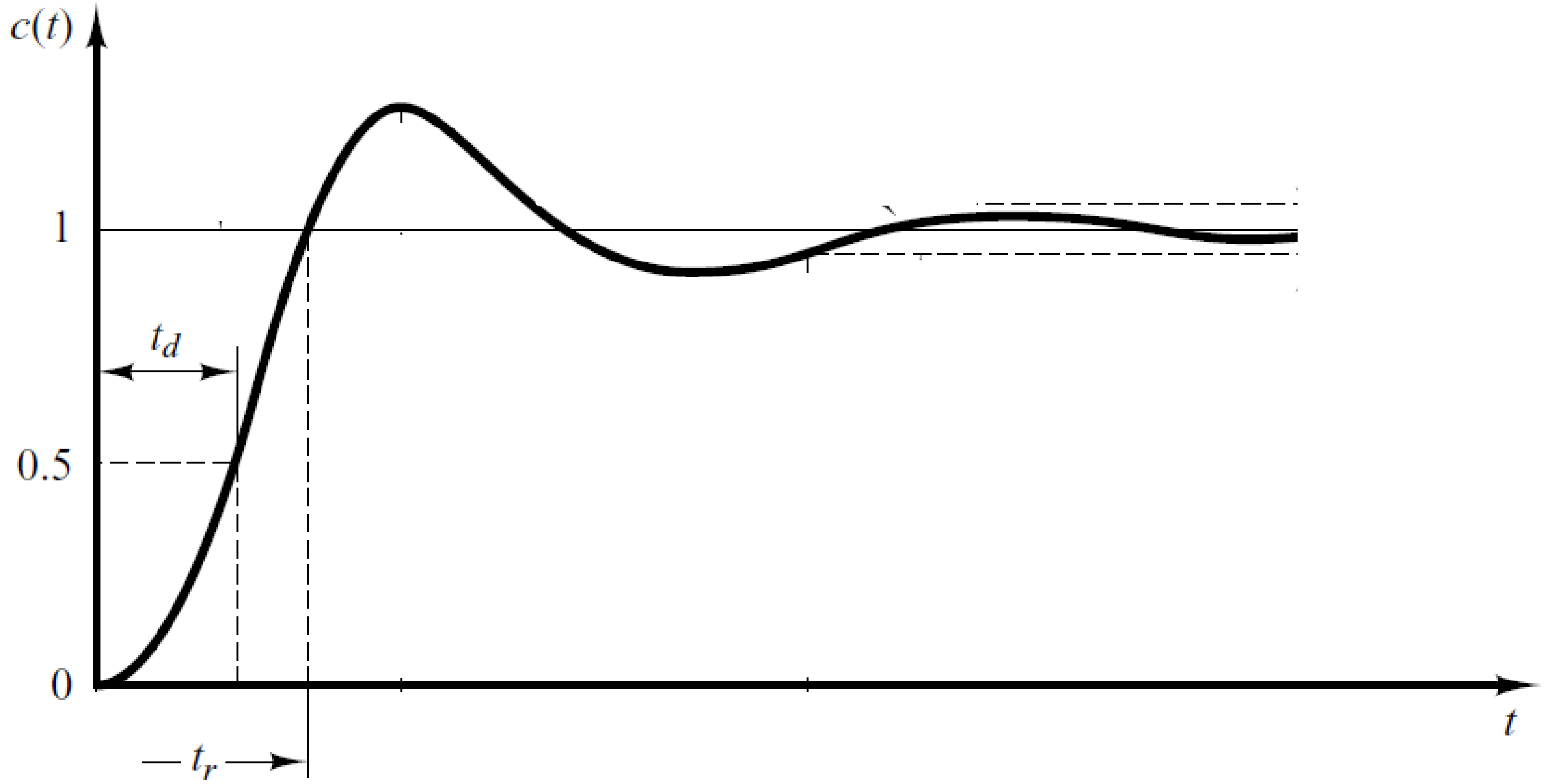
Most systems are underdamped where $0 < \zeta < 1$ and $\omega_n > 0$, these systems are of most interest. The 2nd order system's response due to a unit step input looks like



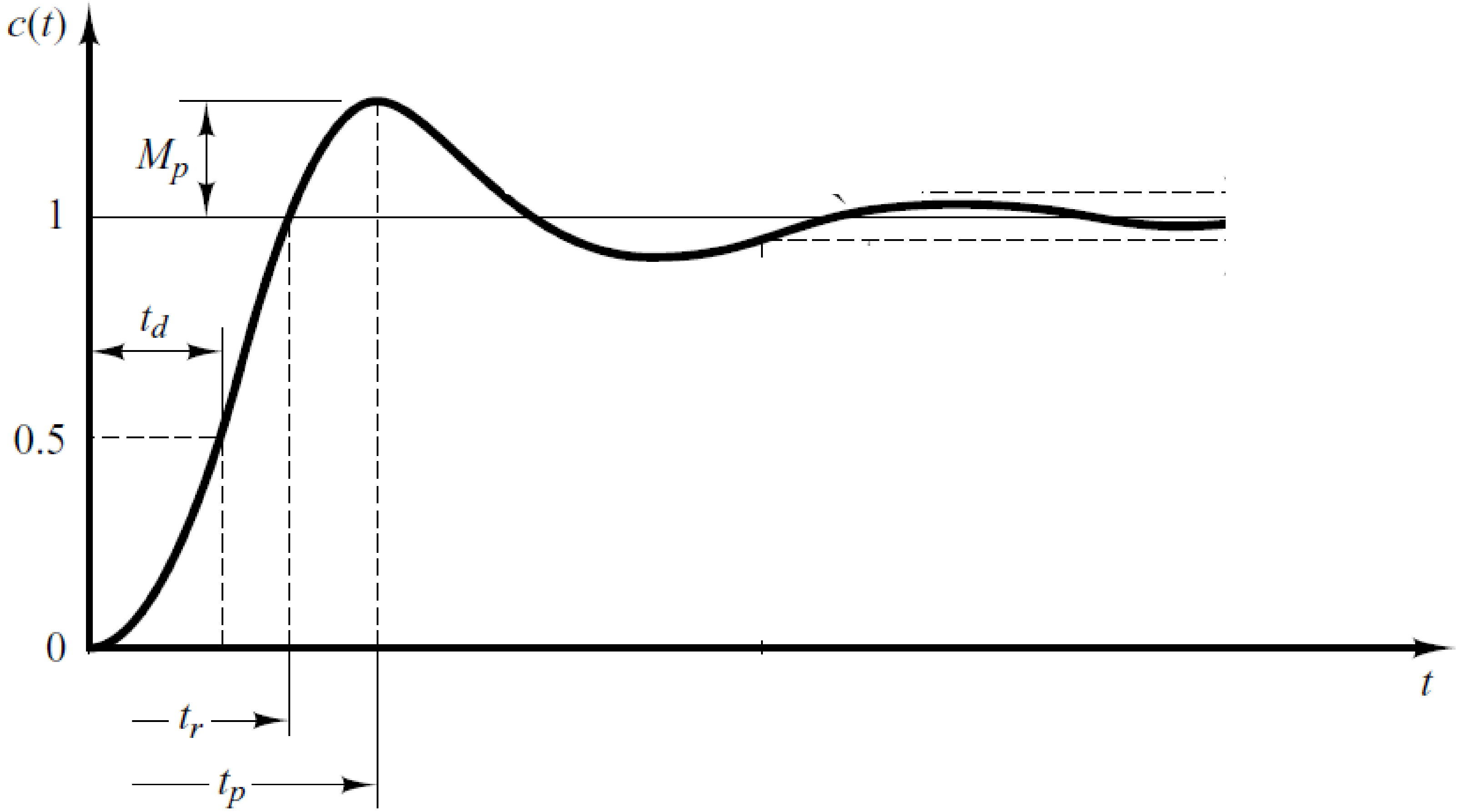
- The delay (t_d) time is the time required for the response to reach half the final value the very first time.



- The rise time is the time required for the response to rise from 10% to 90%, 5% to 95%, or 0% to 100% of its final value.
- For underdamped second order systems, the 0% to 100% rise time is normally used. For overdamped systems, the 10% to 90% rise time is commonly used.



- The peak time is the time required for the response to reach the first peak of the overshoot.



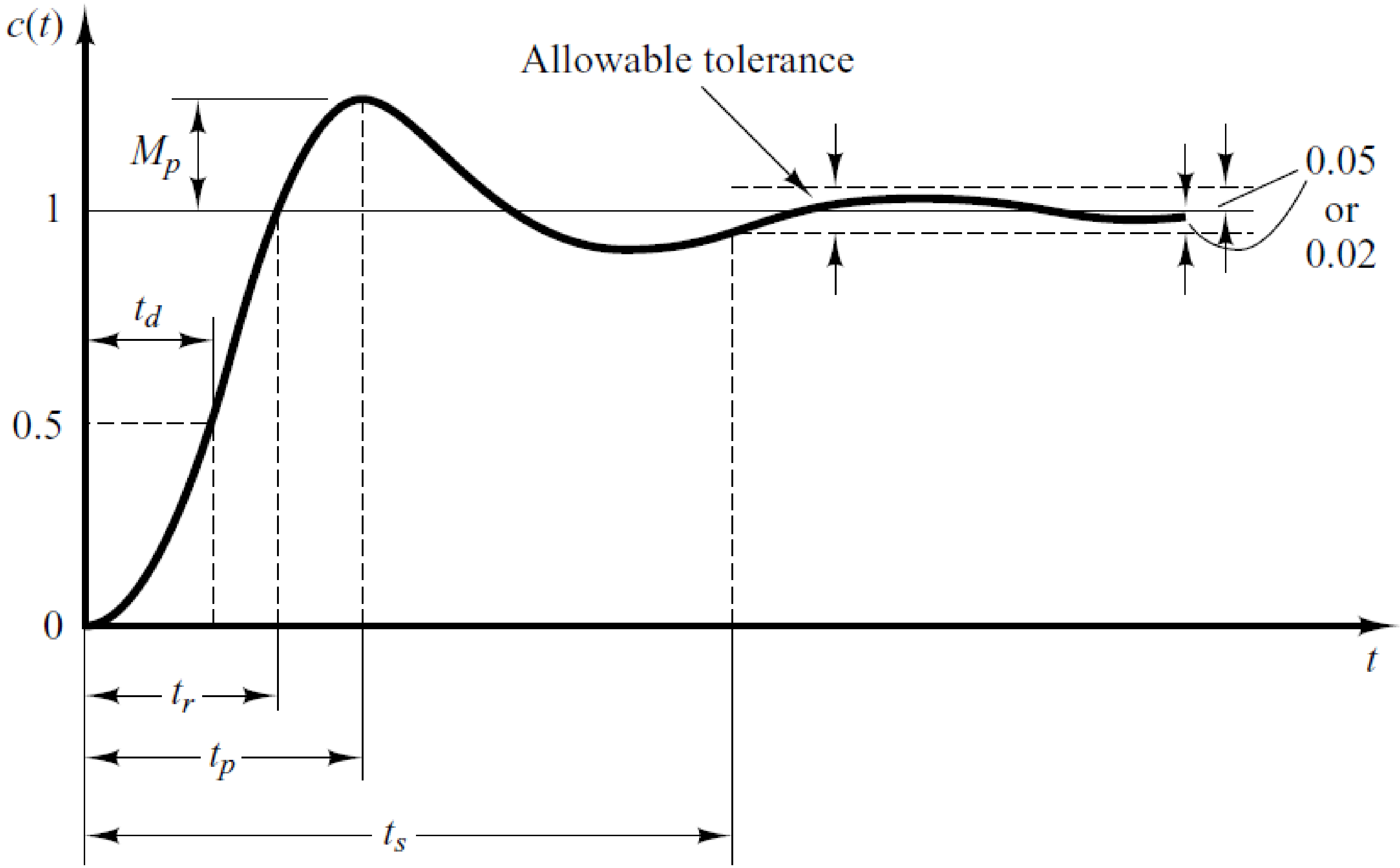
The maximum overshoot is the maximum peak value of the response curve measured from unity. If the final steady-state value of the response differs from unity, then it is common to use the maximum percent overshoot. It is defined by

$$\text{Maximum percent overshoot} = \frac{c(t_p) - c(\infty)}{c(\infty)} \times 100\%$$

The amount of the maximum (percent) overshoot directly indicates the relative stability of the system.

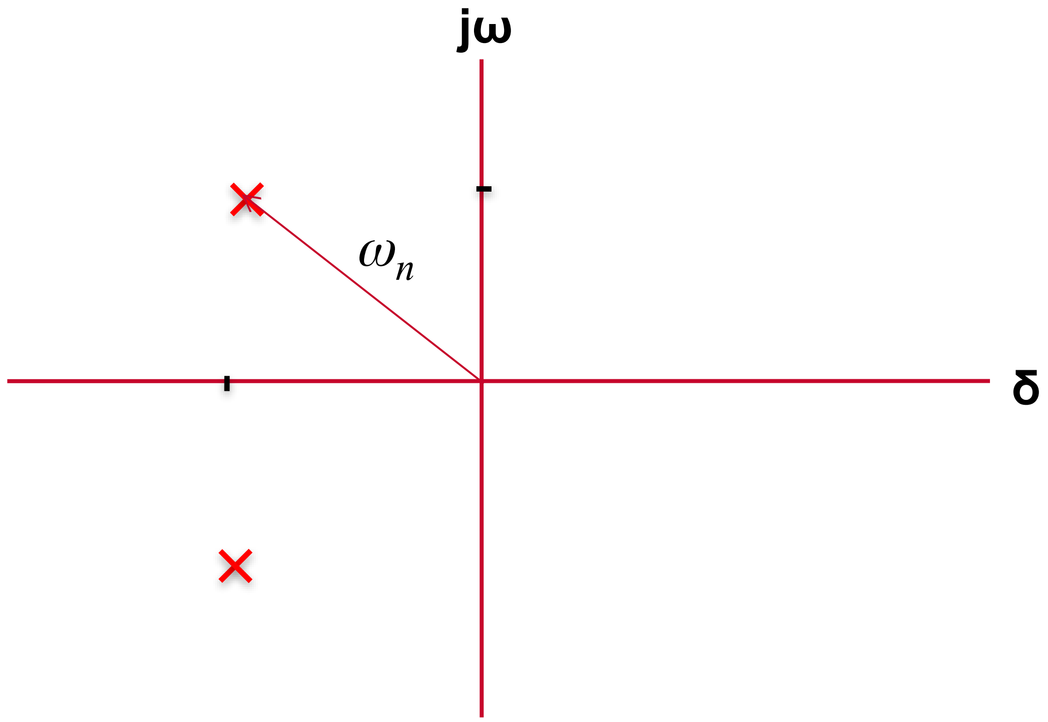
Time-Domain Specification

- The settling time is the time required for the response curve to reach and stay within a range about the final value of size specified by absolute percentage of the final value (usually 2% or 5%).



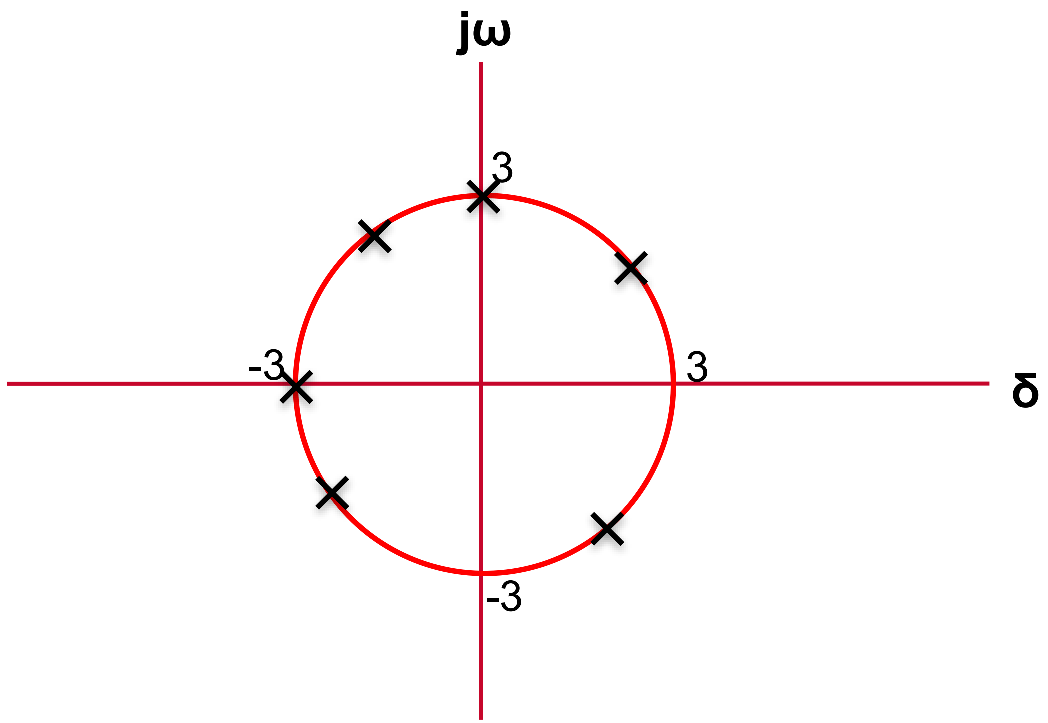
Natural Undamped Frequency.

- Distance from the origin of s-plane to pole is natural undamped frequency in rad/sec.



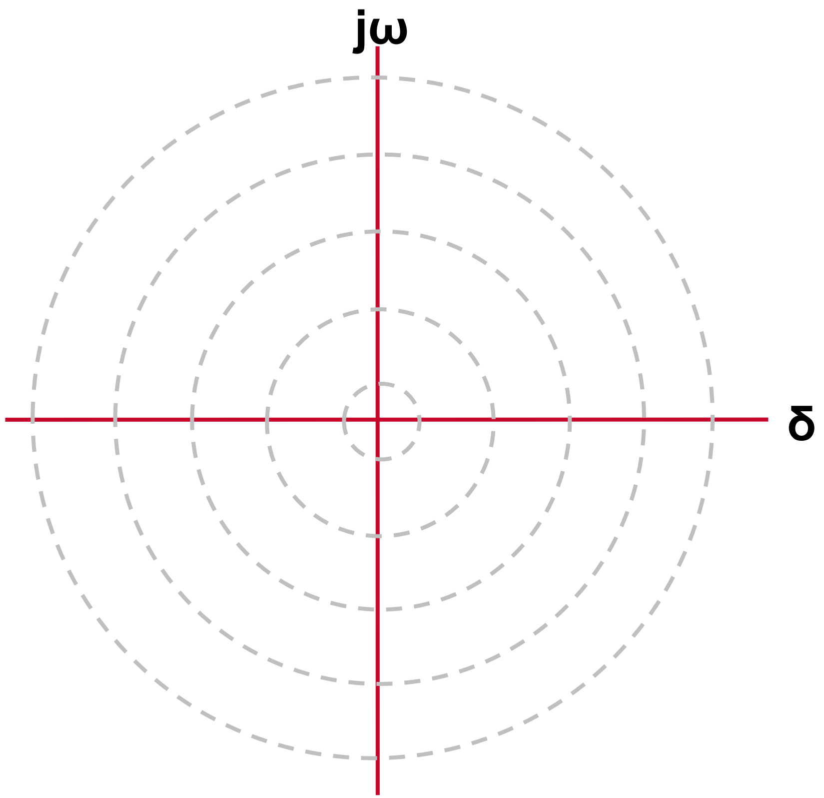
Using the S-Plane

- Let us draw a circle of radius 3 in s-plane.
- If a pole is located anywhere on the circumference of the circle the natural undamped frequency would be *3 rad/sec*.



Using the S-Plane

- Therefore the s-plane is divided into Constant Natural Undamped Frequency (ω_n) Circles.

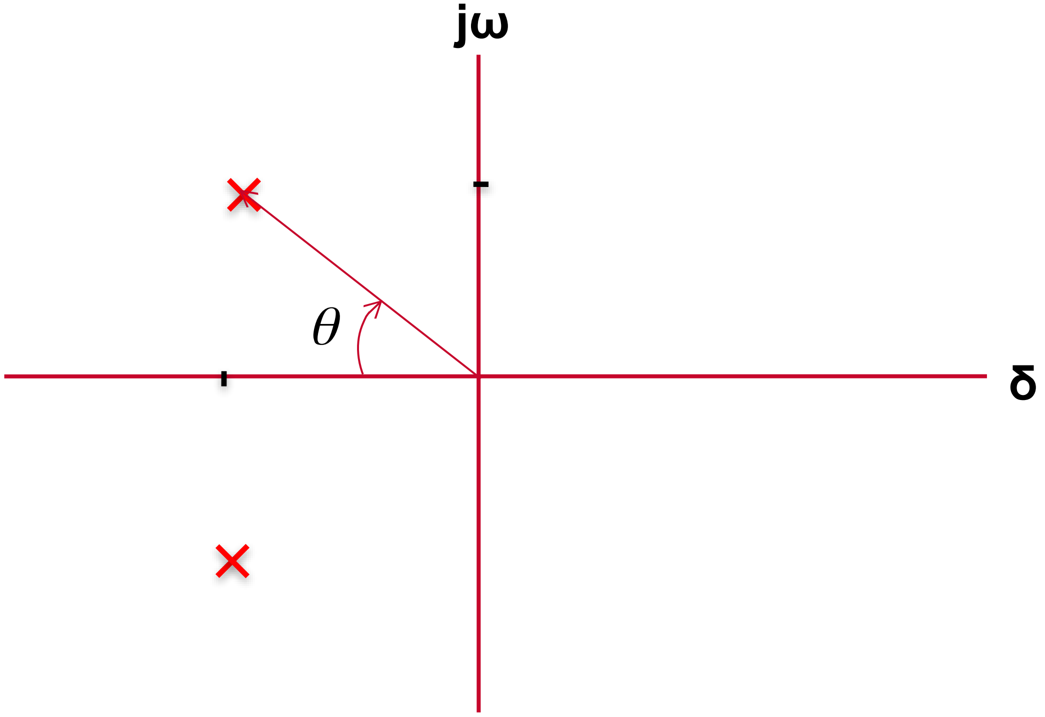


Using the S-Plane

- Damping ratio.

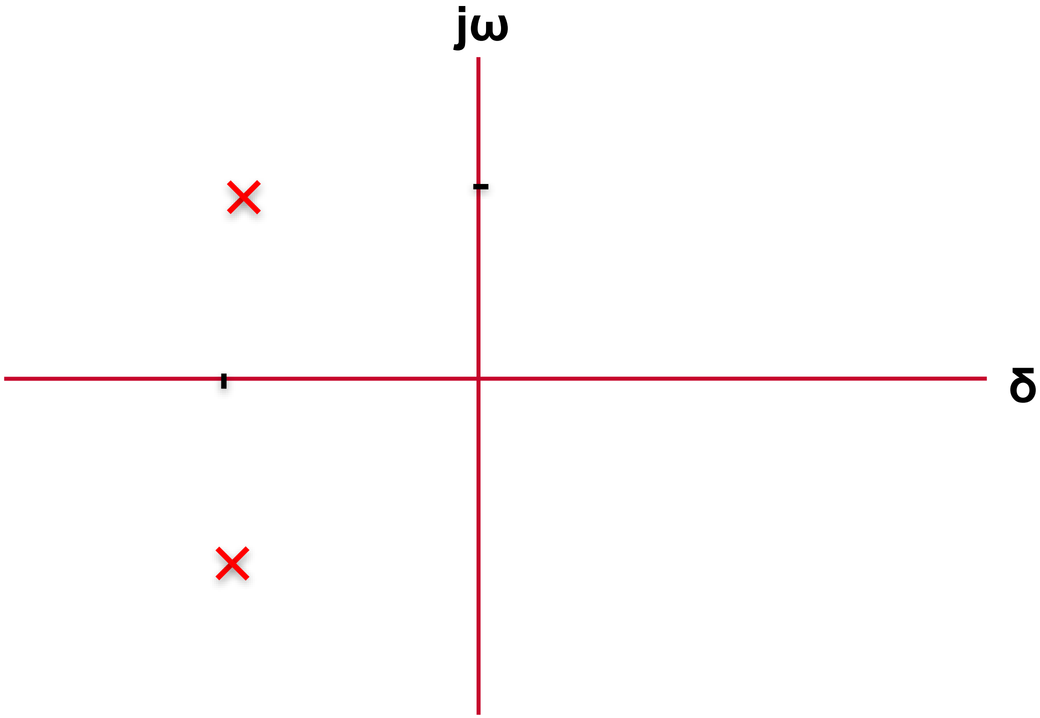
- Cosine of the angle between vector connecting origin and pole and –ve real axis yields damping ratio.

$$\zeta = \cos\theta$$



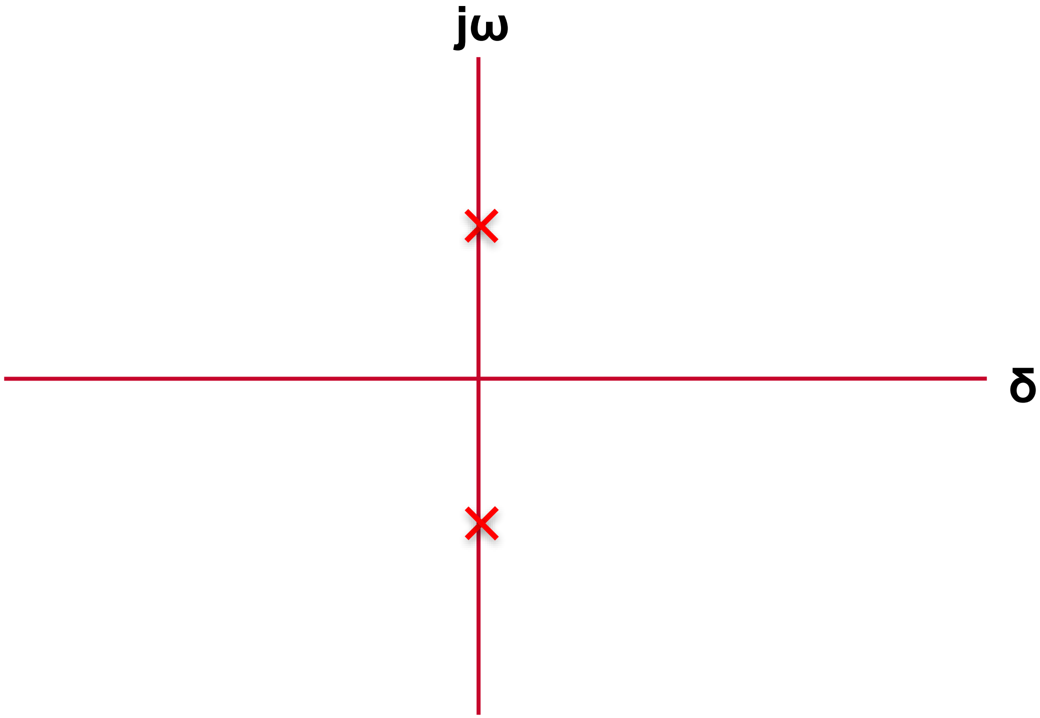
Using the S-Plane

- For Underdamped system $0^\circ < \theta < 90^\circ$ therefore, $0 < \zeta < 1$



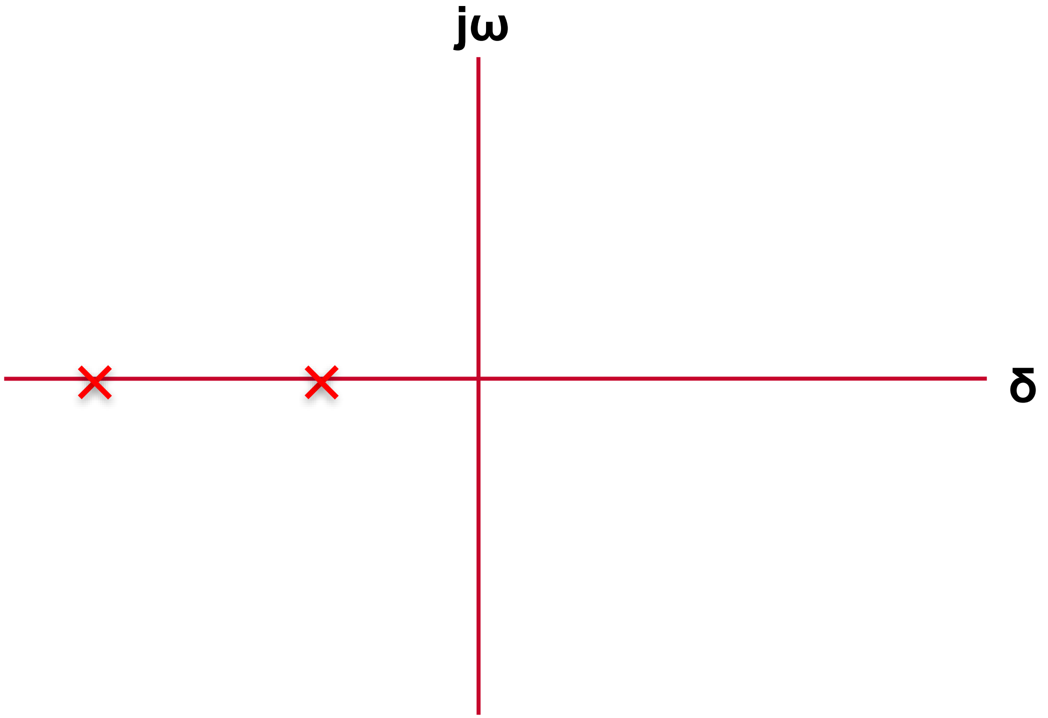
Using the S-Plane

- For Undamped system $\theta = 90^\circ$ therefore, $\zeta = 0$



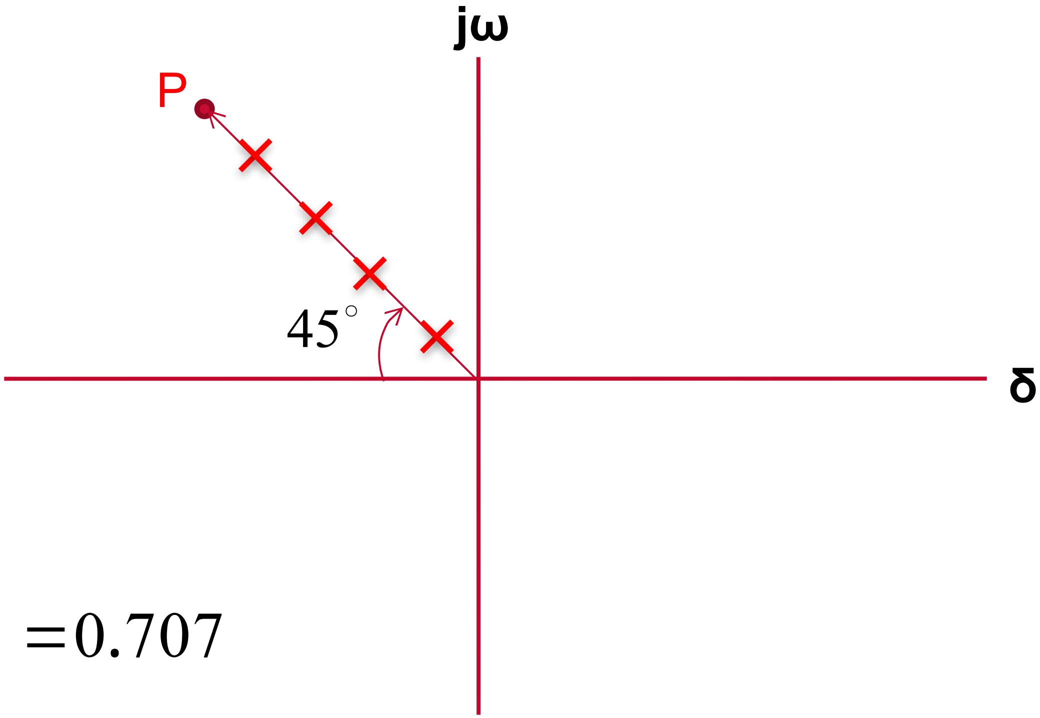
Using the S-Plane

- For overdamped and critically damped systems $\theta = 0^\circ$ therefore, $\zeta = 1$



Using the S-Plane

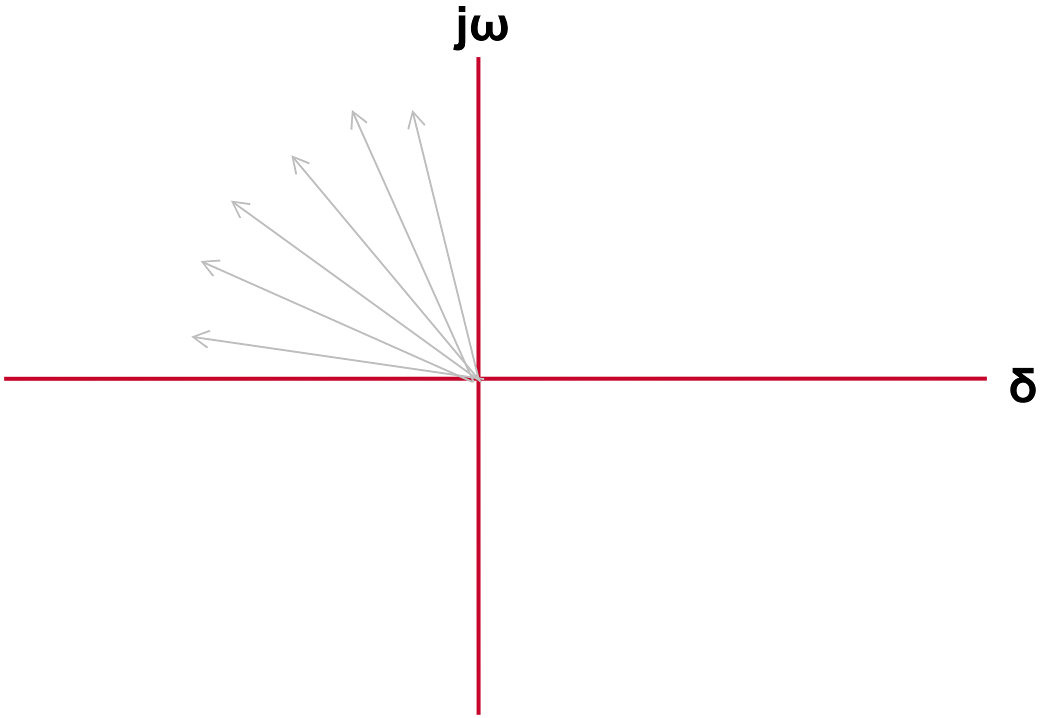
- Draw a vector connecting origin of s-plane and some point P.



$$\zeta = \cos 45^\circ = 0.707$$

Using the S-Plane

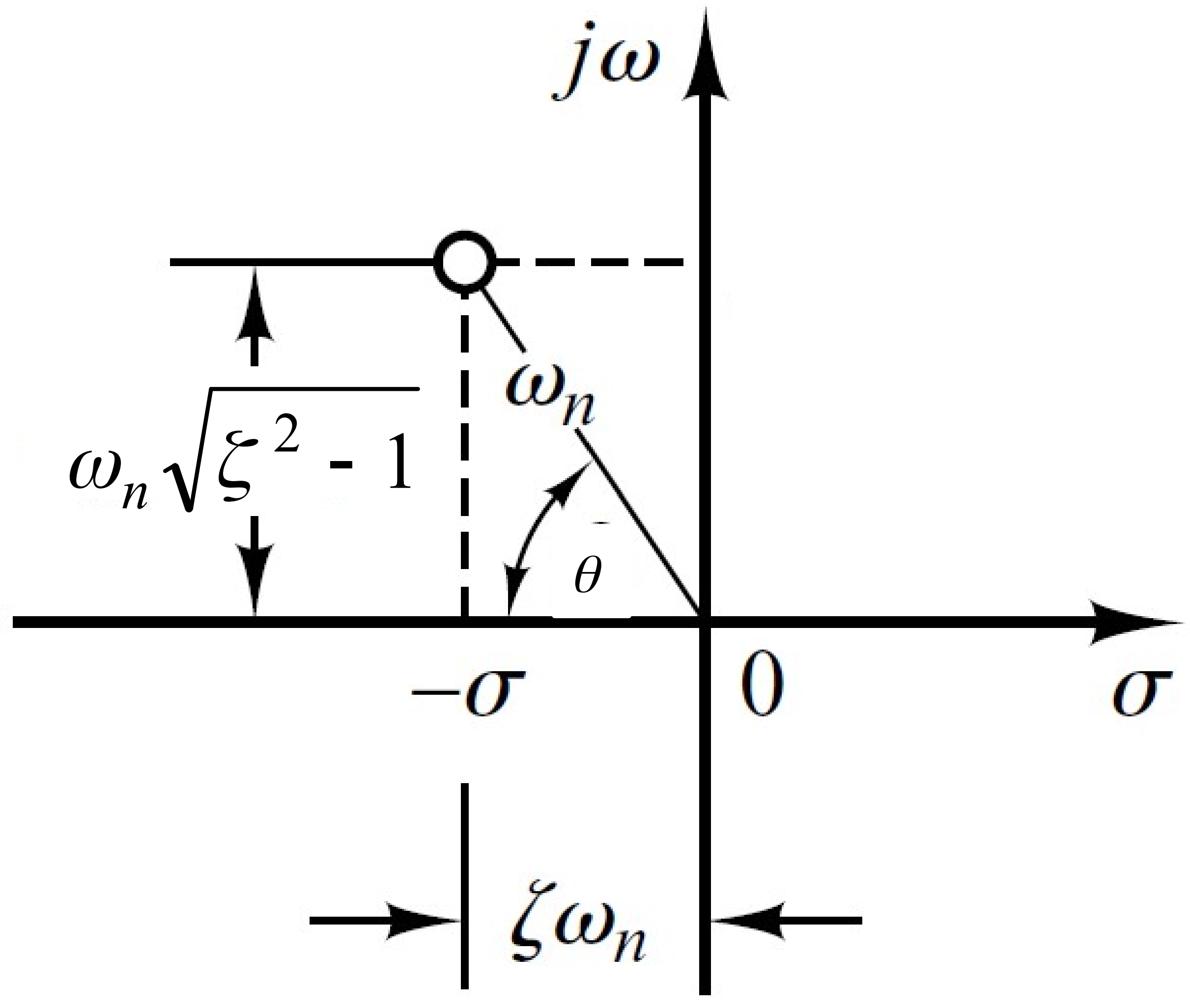
- Therefore, s-plane is divided into sections of constant damping ratio lines.



Using the S-Plane

$$-\omega_n\zeta + \omega_n\sqrt{\zeta^2 - 1}$$

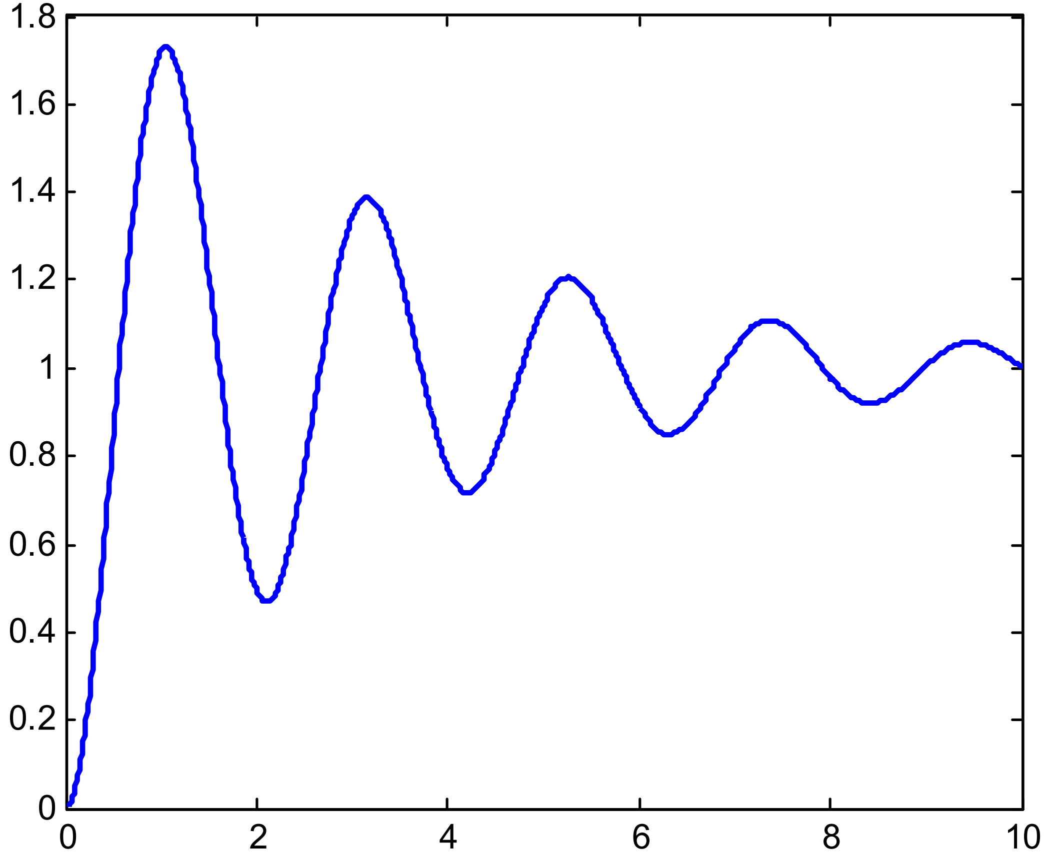
$$-\omega_n\zeta - \omega_n\sqrt{\zeta^2 - 1}$$



Step Response of underdamped System

$$c(t) = 1 - e^{-\zeta\omega_n t}\left[\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right]$$

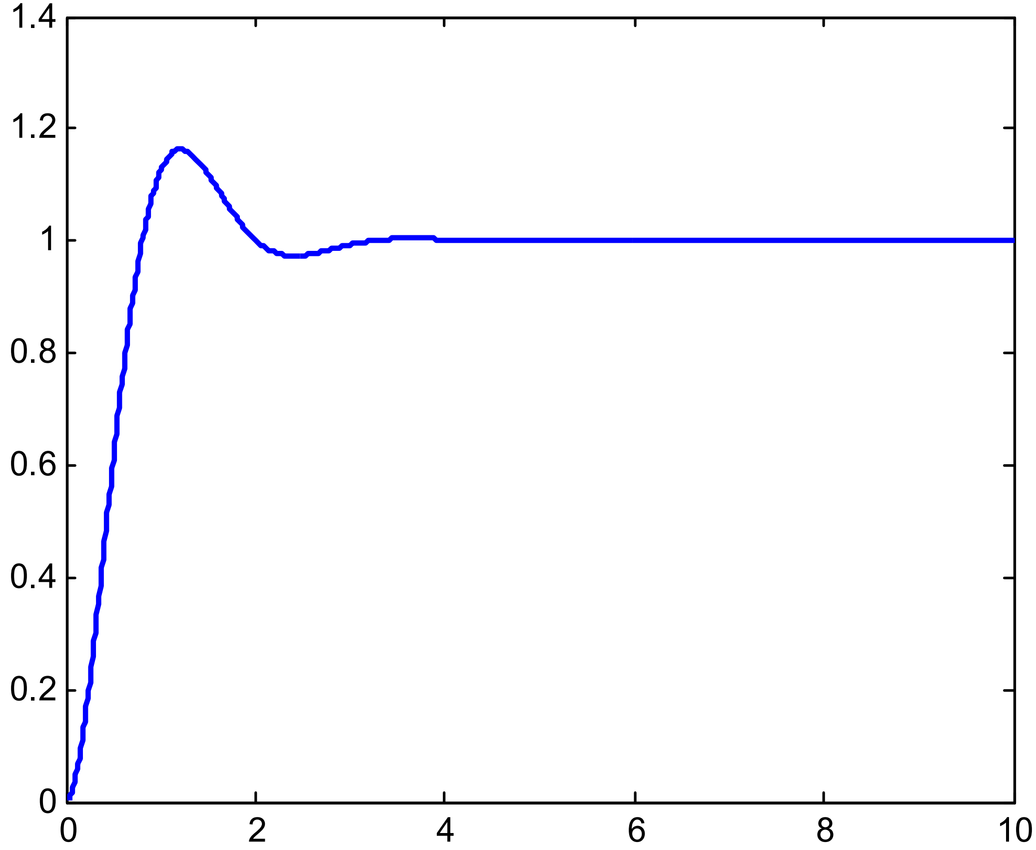
if $\zeta = 0.1$ and $\omega_n = 3\ rad/\sec$



Step Response of underdamped System

$$c(t) = 1 - e^{-\zeta\omega_n t}\left[\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right]$$

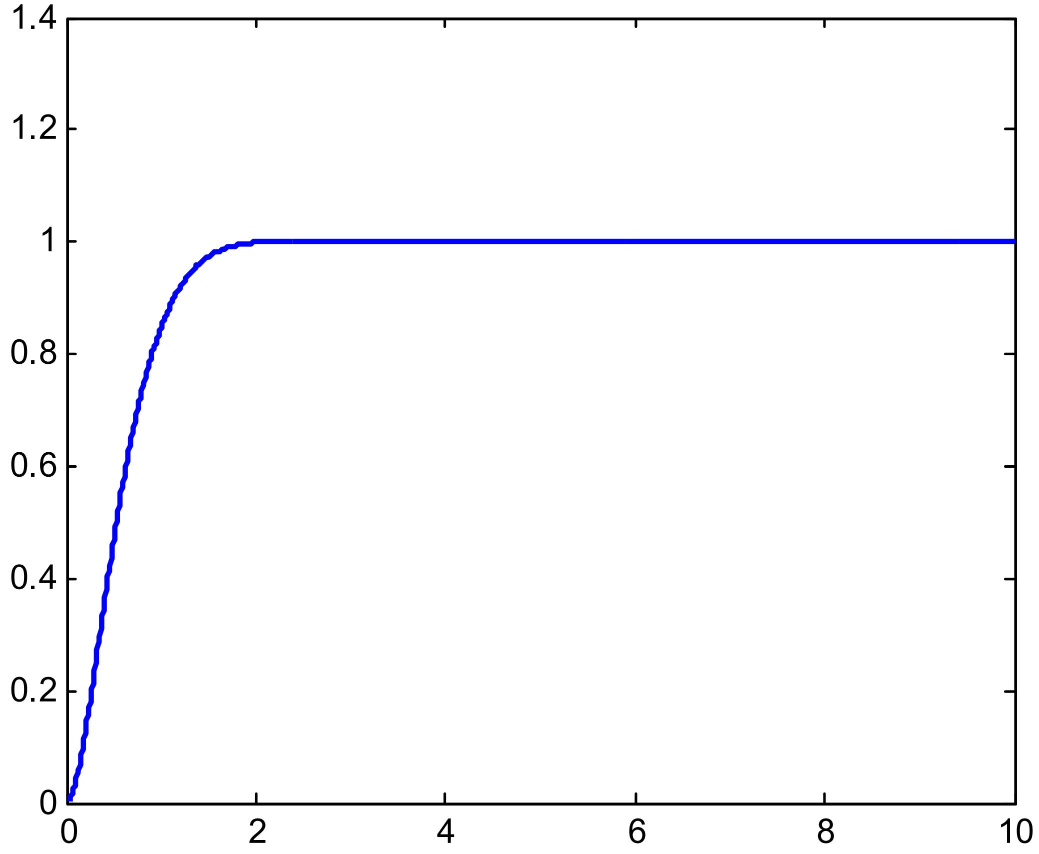
if $\zeta = 0.5$ and $\omega_n = 3 \; rad/\sec$



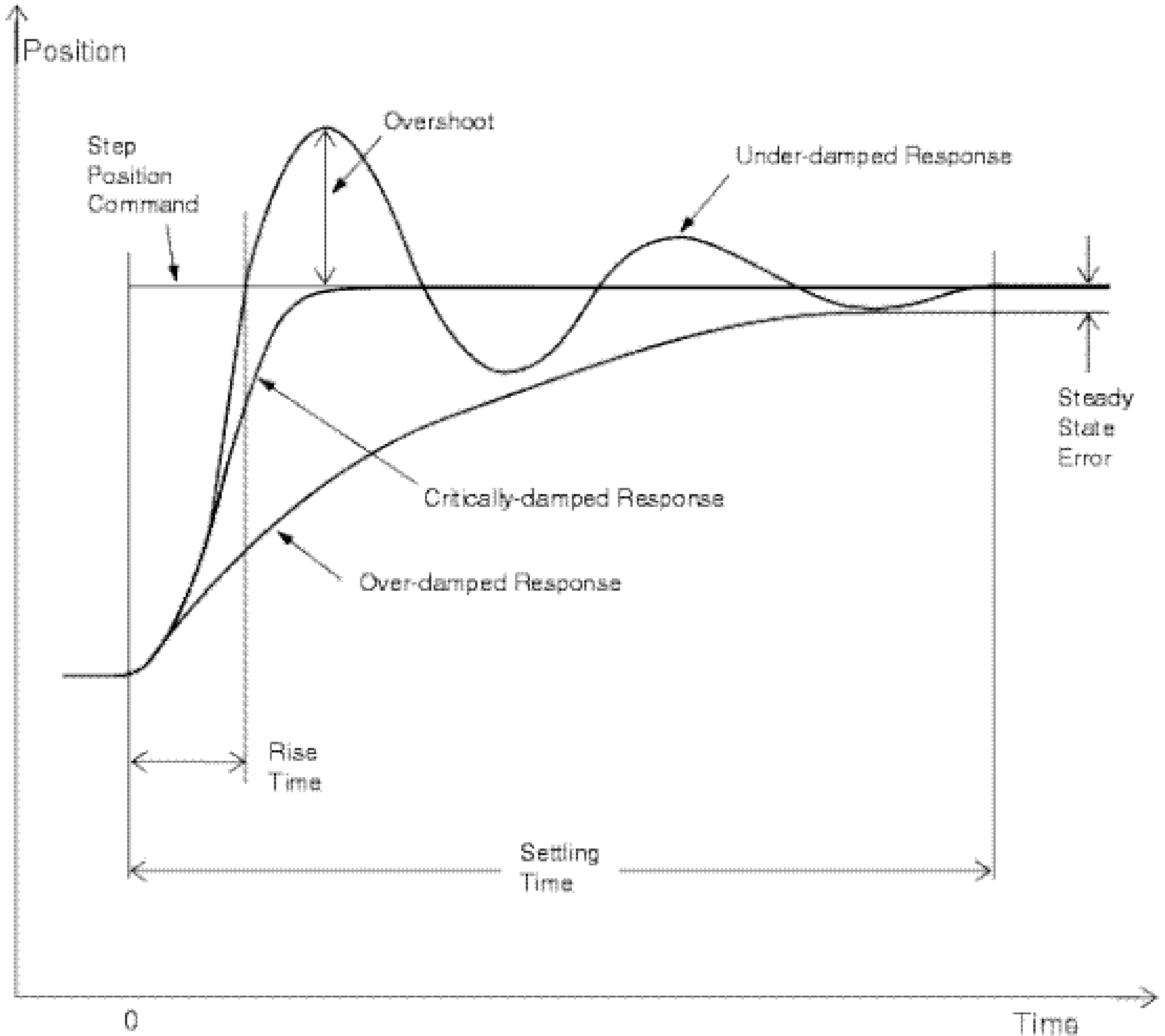
Step Response of underdamped System

$$c(t) = 1 - e^{-\zeta\omega_n t}\left[\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right]$$

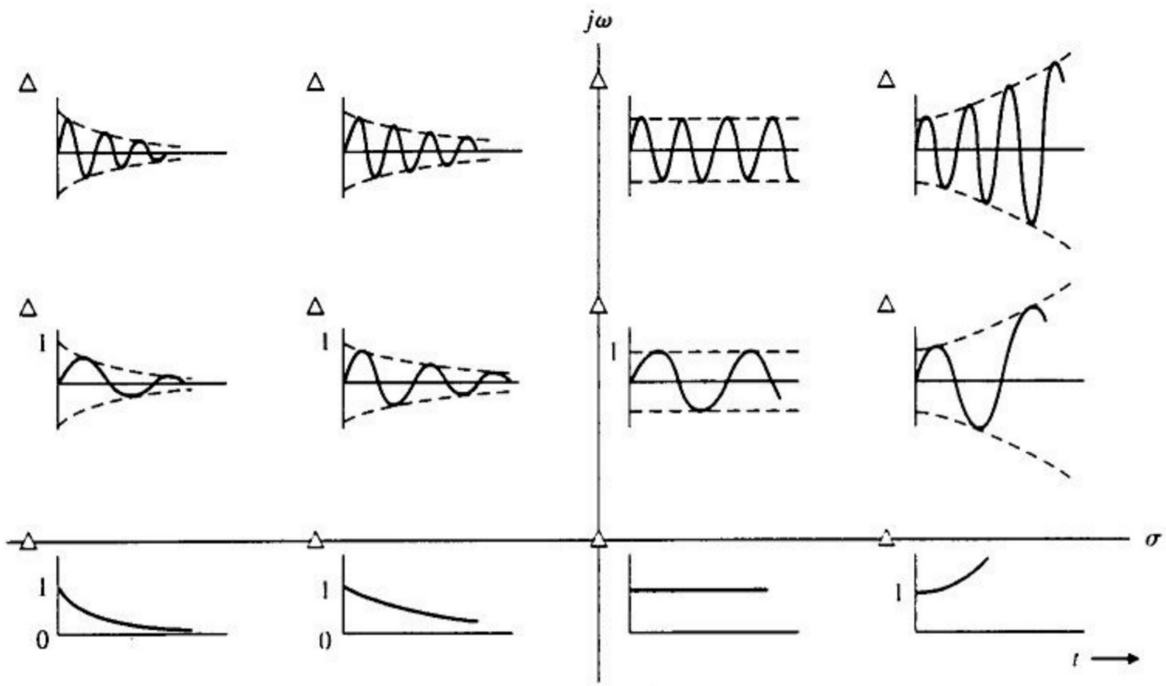
if $\zeta = 0.9$ and $\omega_n = 3 \ rad/\text{sec}$



Other Step Responses



Step Responses by Pole Positions



Higher Frequency Oscillation

Lower Frequency Oscillation

No Oscillation

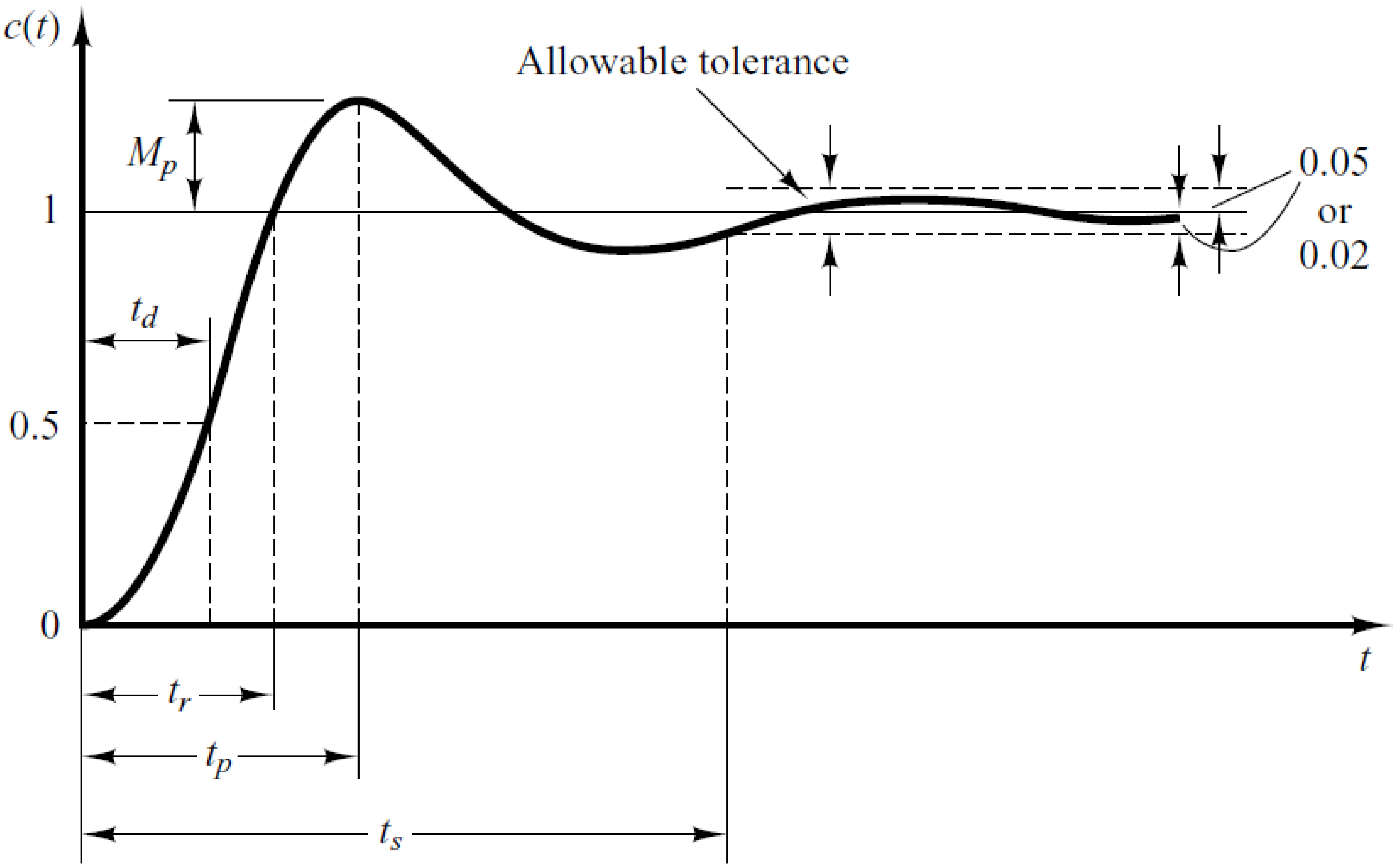
Settling Amplitude

Stable Amplitude

Increasing Amplitude

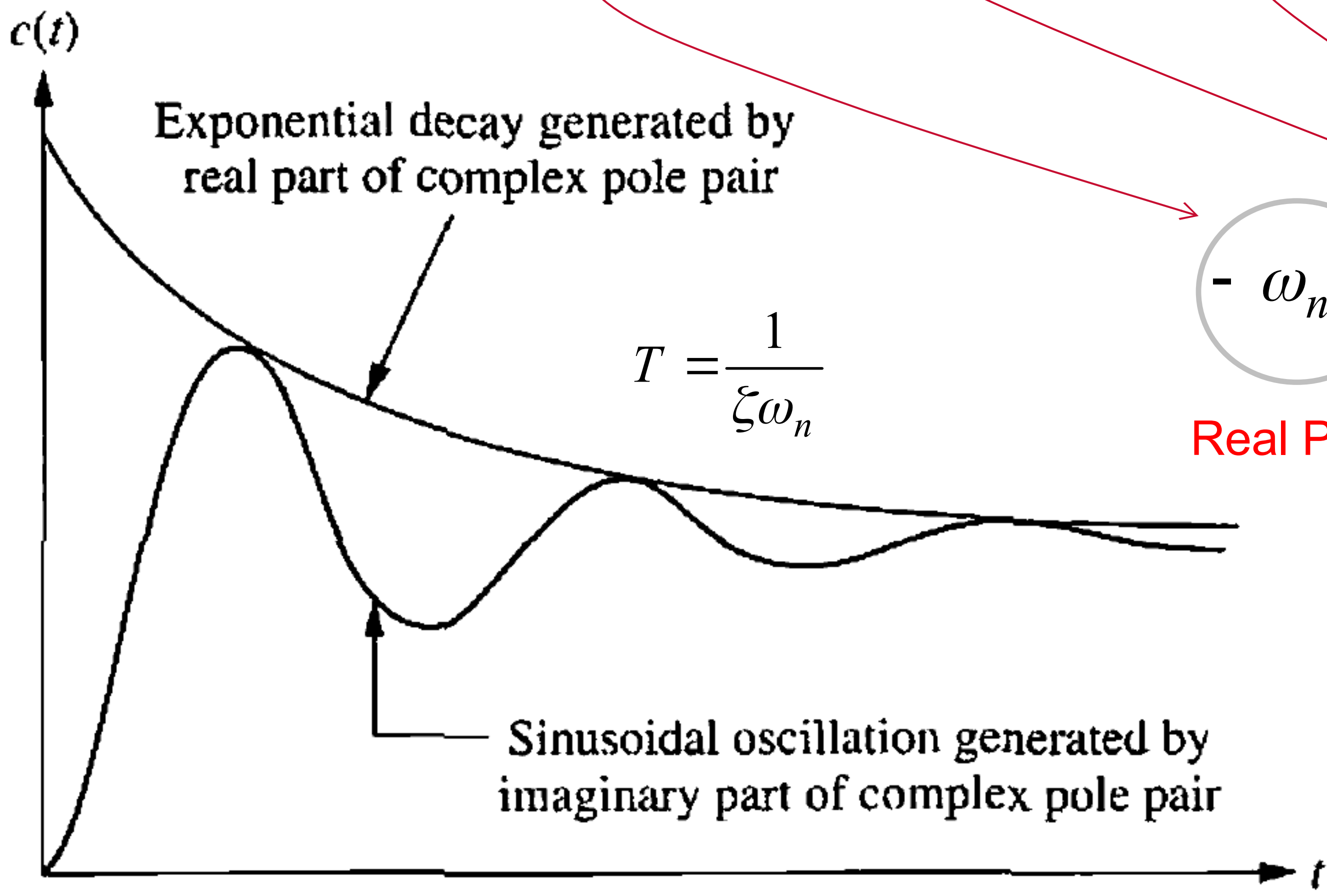
Oscillations Mirrored (+/- Nature of poles)

Time Domain Specifications (Settling Time)



Time Domain Specifications (Settling Time)

$$c(t) = 1 - e^{-\zeta\omega_n t}\left[\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right]$$

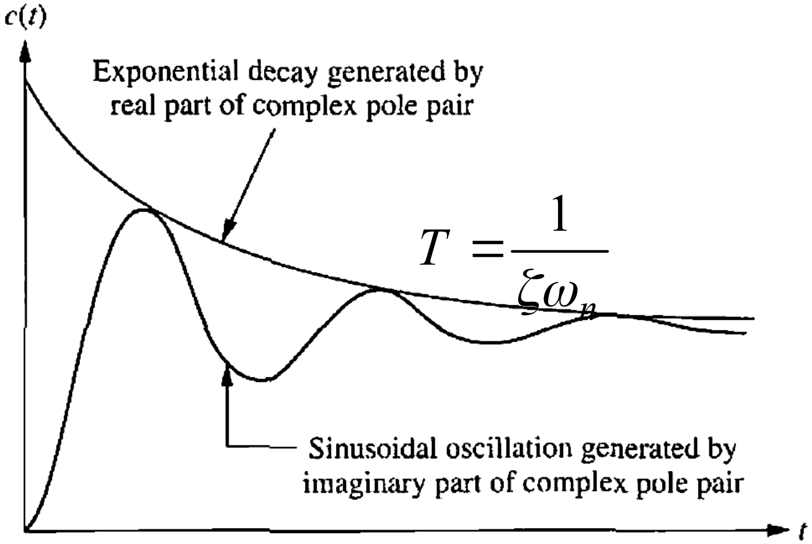


$$-\omega_n\zeta \pm \omega_n\sqrt{\zeta^2 - 1}$$

Real Part   Imaginary Part

Time Domain Specifications (Settling Time)

- Settling time (2%) criterion
    - Time consumed in exponential decay up to 98% of the input.



$$t_s = 4T = \frac{4}{\zeta\omega_n}$$

- Settling time (5%) criterion
    - Time consumed in exponential decay up to 95% of the input.

$$t_s = 3T = \frac{3}{\zeta\omega_n}$$

Summary of Time Domain Specifications

Rise Time

$$t_r = \frac{\pi - \theta}{\omega_d} = \frac{\pi - \theta}{\omega_n \sqrt{1 - \zeta^2}}$$

Peak Time

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

Settling Time (2%)

$$t_s = 4T = \frac{4}{\zeta \omega_n}$$

$$t_s = 3T = \frac{3}{\zeta \omega_n}$$

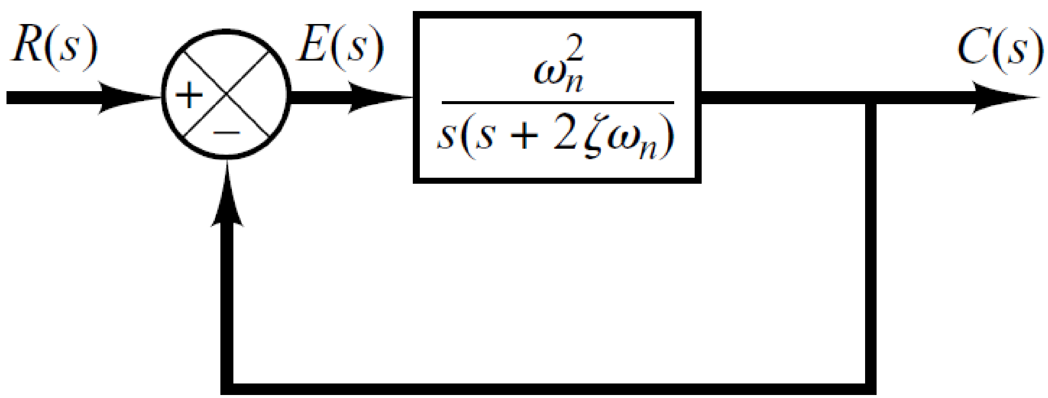
Settling Time (5%)

Maximum Overshoot

$$M_p = e^{-\frac{\pi \zeta}{\sqrt{1 - \zeta^2}}} \times 100$$

Example

Consider the system shown in following figure, where damping ratio is 0.6 and natural undamped frequency is 5 rad/sec. Obtain the rise time t_r, peak time t_p, maximum overshoot M_p, and settling time 2% and 5% criterion t_s when the system is subjected to a unit-step input.



Example

Rise Time

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$t_r = \frac{3.141 - \theta}{\omega_n \sqrt{1 - \zeta^2}}$$

$$\theta = cos^{-1}\zeta = \tan^{-1}(\text{¿}\, \frac{\omega_n \sqrt{1-\zeta^2}}{\zeta\,\omega_n}) = 0.93 \ \text{rad}\, \text{¿}$$

$$t_r = \frac{3.141 - 0.93}{5\sqrt{1 - 0.6^2}} = 0.55s$$

Example

Peak Time

$$t_p = \frac{\pi}{\omega_d}$$

$$t_p = \frac{3.141}{4} = 0.785s$$

Settling Time (2%)

$$t_s = \frac{4}{\zeta\omega_n}$$

$$t_s = \frac{4}{0.6 \times 5} = 1.33s$$

Settling Time (5%)

$$t_s = \frac{3}{\zeta\omega_n}$$

$$t_s = \frac{3}{0.6 \times 5} = 1s$$

Example

Maximum Overshoot

$$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100$$

$$M_p = e^{-\frac{3.141\times 0.6}{\sqrt{1-0.6^2}}} \times 100 = 0.094 \times 100 = 9.4\%$$

PID Control

PID-Control – Introduction

The PID-controller is a “combination” of three different actions:

*** The “P-Action”**

*** The “I-Action”**

*** The “D-Action”**

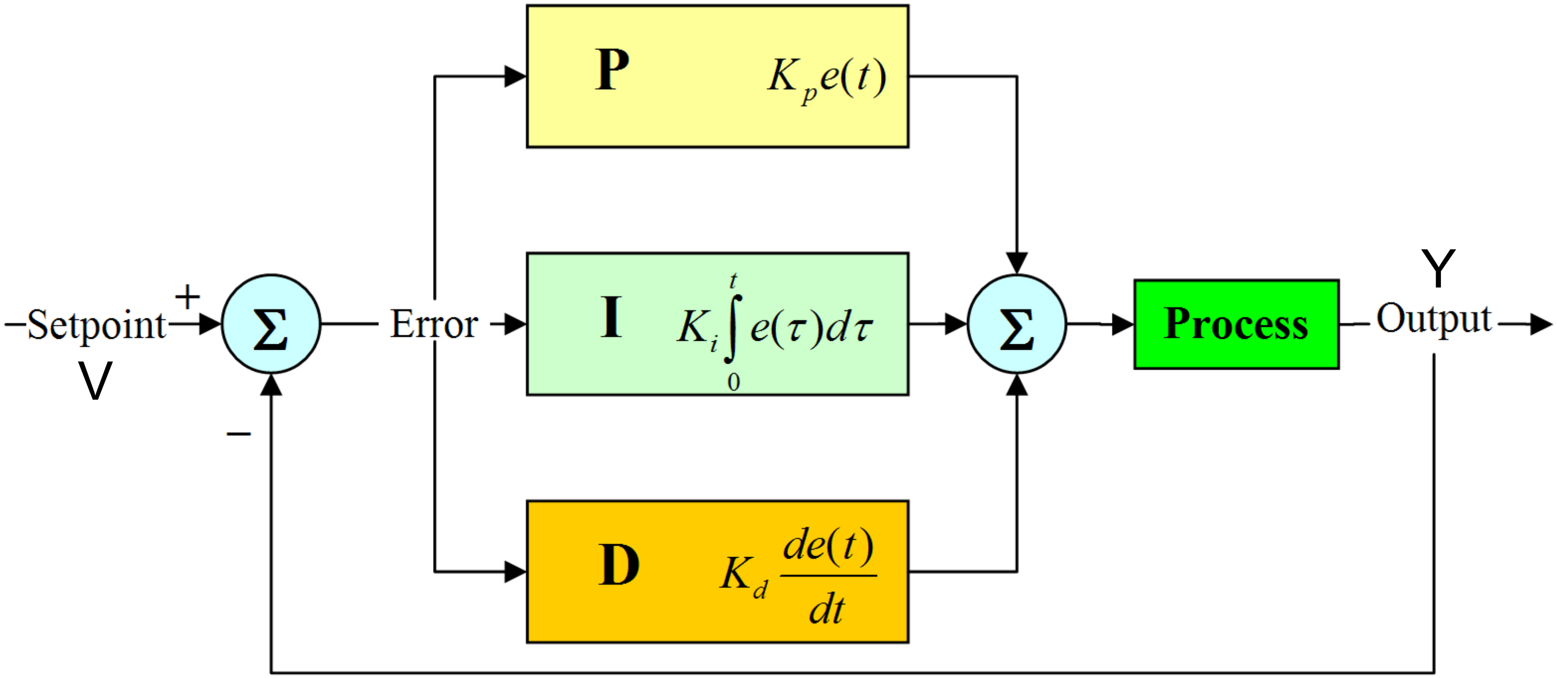
Industrial controllers come with different classifications

- Proportional controllers
- Integral controllers
- Proportional-plus-integral controllers
- Proportional-plus-derivative controllers
- *Proportional-plus-integral-plus-derivative controllers*

This Lecture will explain these actions “one-by-one”, so that each action will be easy to understand

Proportional-Integral-Derivative (PID) Control

The PID-controller combines the three actions in parallel



At any given time the Error signal will be: $e(t) = v - y(t)$

Proportional-Integral-Derivative (PID) Control

Hence the transfer function of the controller is the sum of the three elements

$$u(t) = kp\, e(t) + ki \int e(t)\, dt \;\; + kd\, de(t)/dt$$

The gain k can be weighted to determine the relative influence of each control type.

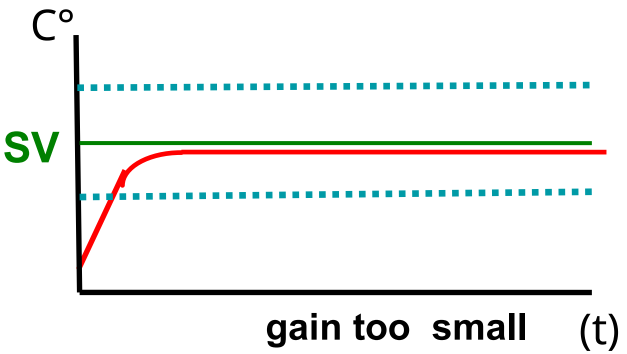
Proportional-Integral-Derivative (PID) Control

	Good	Bad
Proportional	Reduces steady-state error	Too much causes instability
Integral	Eliminates steady-state error	Too much causes instability
Derivative	Adds damping reduces overshoot	Amplifies noise

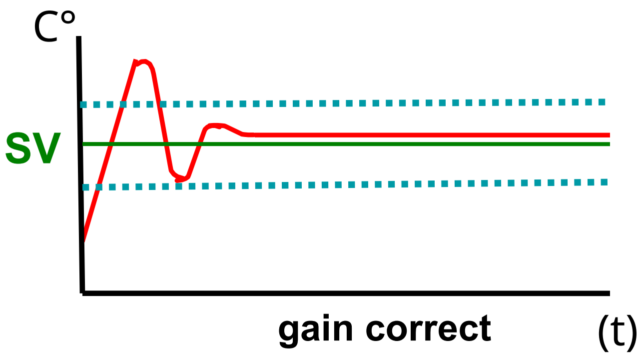
Proportional Control

With a low gain the Setpoint temperature will never be reached!

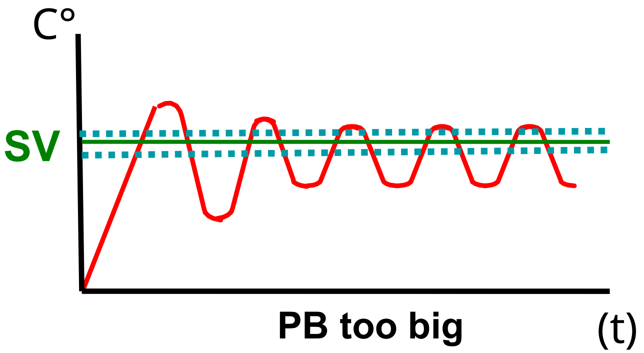
This will create a large offset from the Set Point!



A correctly sized P gain results in an Overshoot, followed by an Undershoot and than Stabilization, with a small offset near the Set Point.

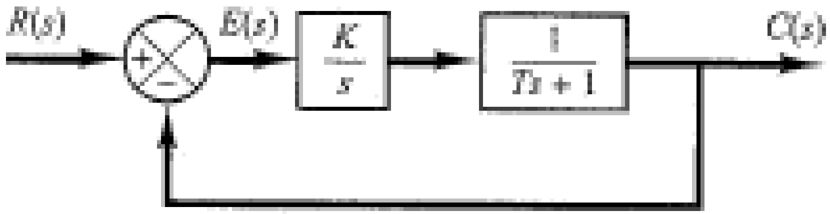


A Proportional gain that is too high causes hunting! The TC will than behave like an ON/OFF controller!

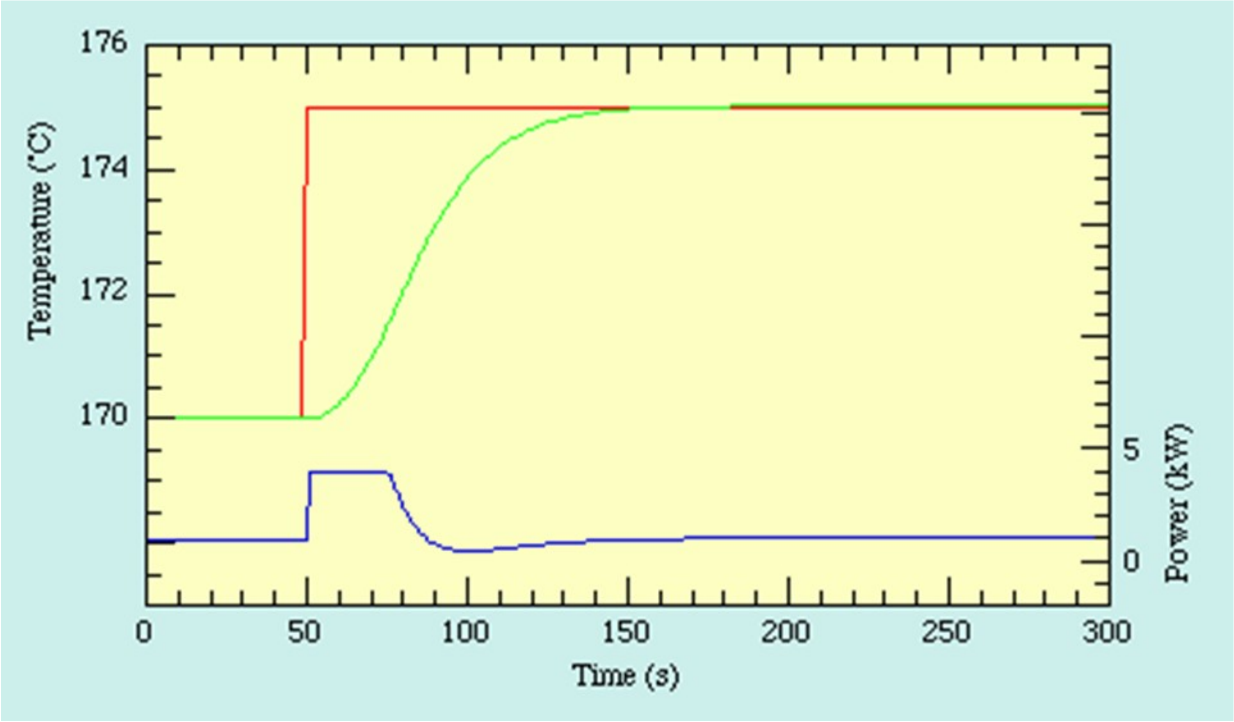


The right setting of the proportional gain is very important

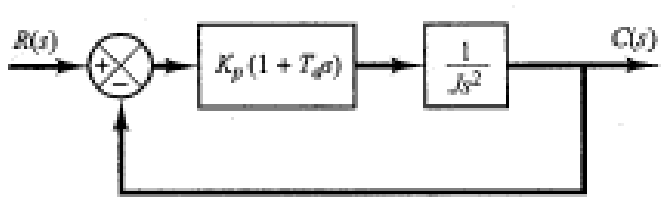
Integral Control of Systems.



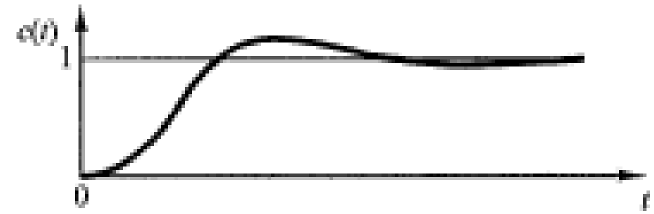
Integral control of the system eliminates the steady-state   error in the response to the step input..



Derivative Control



(a)

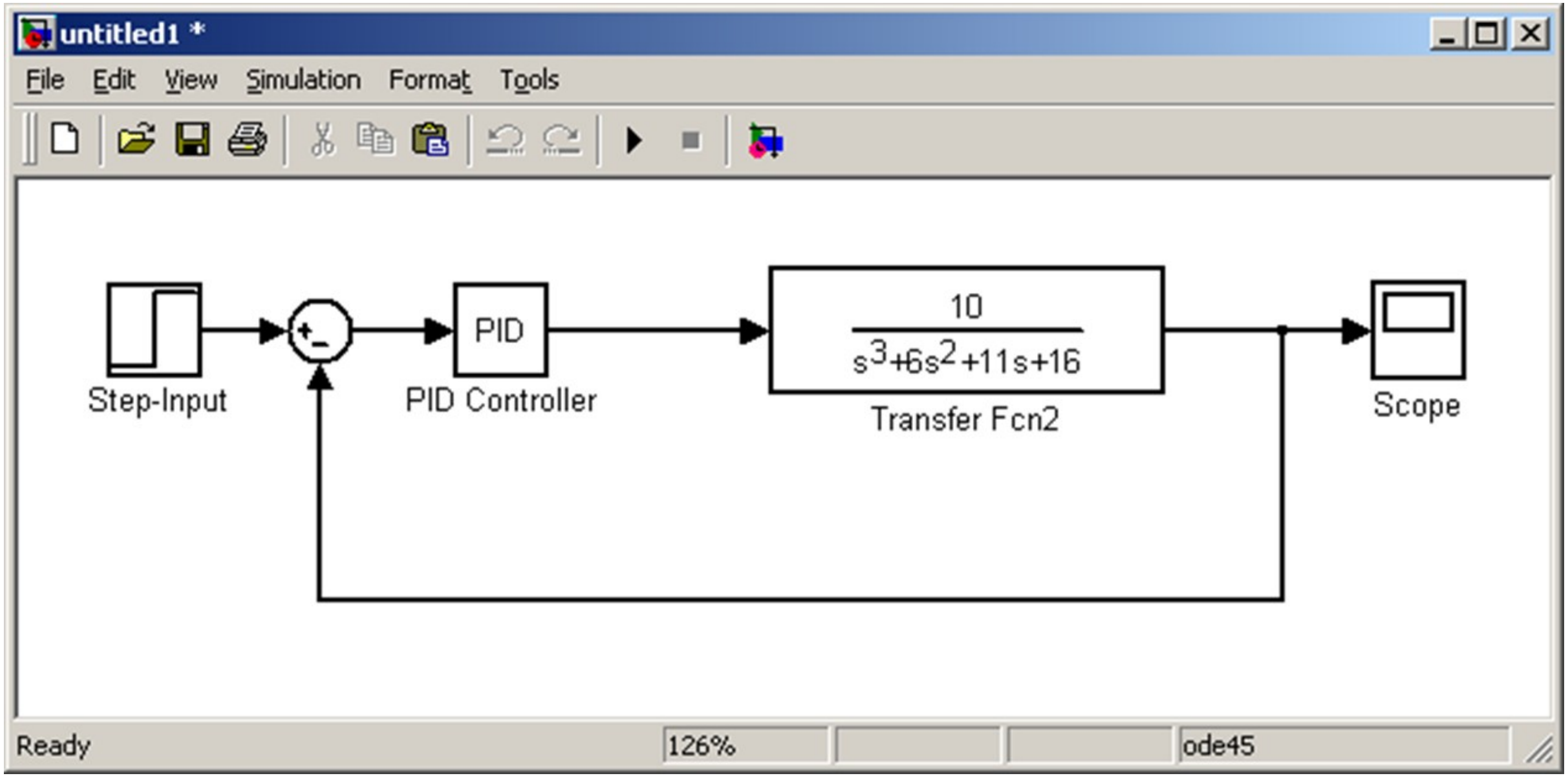


(b)

$$\frac{C(s)}{R(s)} = \frac{K_p(1 + T_d s)}{Js^2 + K_p T_d s + K_p}$$

Derivative control introduces a damping effect. A typical response curve ***c (t) to a unit step*** input is shown

Simulink Simulation of PID control system



The following additional explanation can also help to understand the actions of the PID-controller:

- The "P-Action" deals with the "present"
  Depending on the deviation from the Setpoint: more or less Output capacity will be given.

- The "I-Action" deals with the "past"
  If we have been below setpoint: the Output will be increased.
  If we have been above setpoint: the Output will be decreased.

- The "D-Action" deals with the "future"
  If the Temperature is going down: the Output will be increased.
  If the Temperature is going up: the Output will be decreased.

P=Present

I=Past

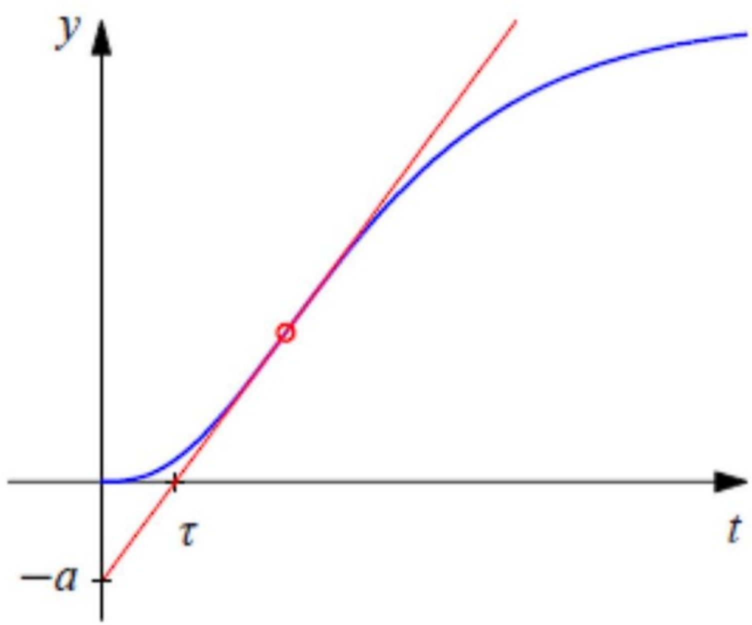
D=Future

This "combination", of "Present + Past + Future", makes it possible to control the application very well.

PID controller tuning- Ziegler-Nichols reaction curve method

Over 60 years ago, Ziegler and Nichols (1942) published a classic paper that introduced the *continuous cycling method* for controller tuning. It is based on the following trial-and-error procedure:

The open loop step response curve of the process is recorded and the steepest tangent to the reaction curve is drawn



PID controller tuning- Ziegler-Nichols reaction curve method

The reaction curve is characterized by two parameters:

- Ƭ intercept of the steepest tangent with the time axis

- a modulus of the intercept of the steepest tangent with the time axis.

Note that Ƭ is an approximation of the time delay of the system and Ƭ/a is the steepest slope of the step response.

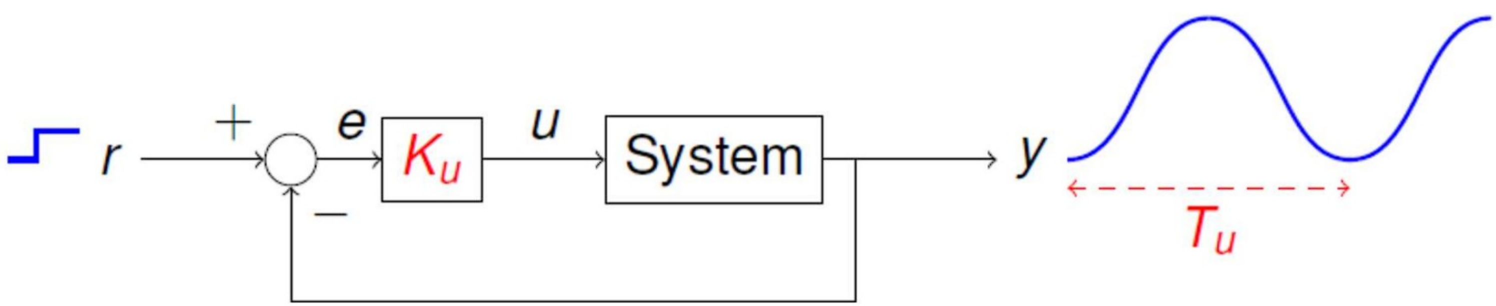
PID controller tuning- Ziegler-Nichols reaction curve method

Table shows the PID tuning rules based on the two reaction curve parameters. The table also shows an estimate of the period of oscillation Tp of underdamped step response the closed loop system.

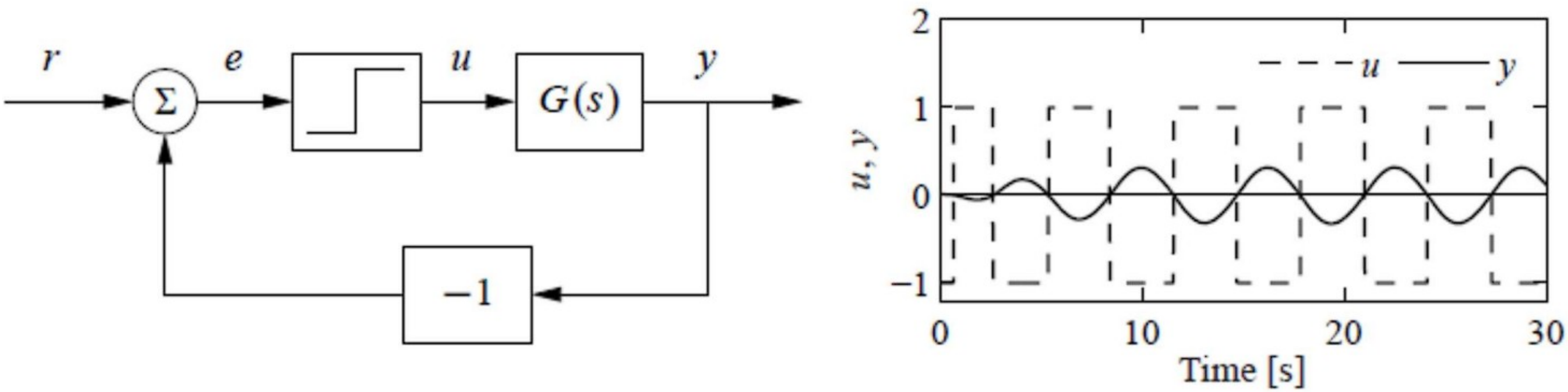
Controller	K_p	T_i	T_d	T_p
P	1/a	-	-	4τ
PI	0.9/a	3τ	-	5.7τ
PID	1.2/a	2τ	0.5τ	3.4τ

Also known as the Ziegler-Nichols frequency response (ZNFR) method.

With the system under proportional-only control, a unit step input is applied to the closed-loop system and the proportional gain is increased until the response is a sustained oscillation



An alternative way to obtain the sustained oscillation required for the Ziegler–Nichols ultimate gain tuning method is to connect the process in a feedback loop with a nonlinear element having a square wave relay function



(a) Relay feedback

(b) Oscillatory response

PID controller tuning- Ziegler-Nichols ultimate gain method

In the steady state, the process input u (i.e., relay output) and the process output y will oscillate with the same ultimate period T_u. However, the process input u and output y will be out of phase with the process output lagging by 180°, as shown in ***Figure***

The ultimate gain is estimated from

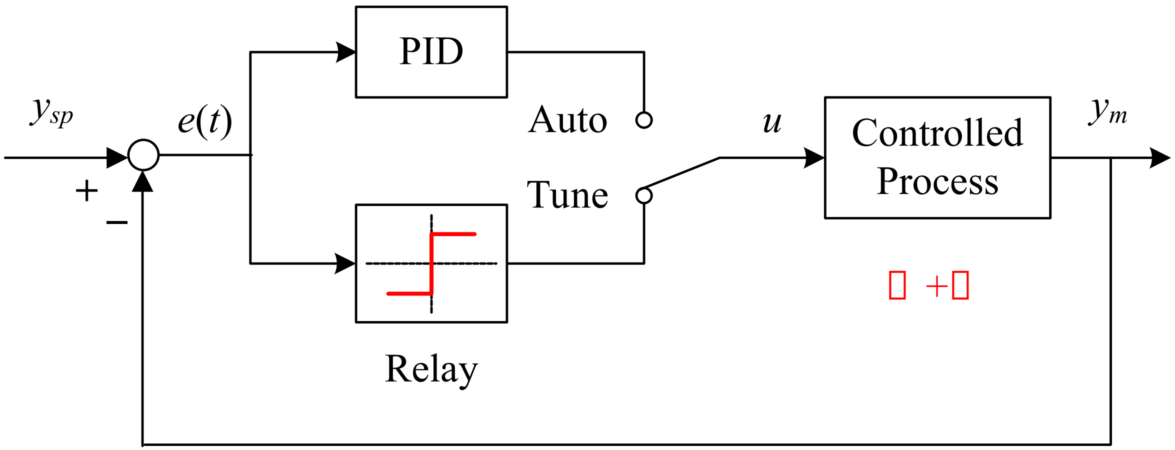
$$K_u = \frac{4d}{\pi a}$$

where *d* is the amplitude of the relay and *a* is the amplitude of the process output. The tuning rules in Table give the PID controller gains.

Ziegler-Nichols Table of Settings

Control Type	K_P	K_I	K_D
P	$0.50K_U$	—	—
PI	$0.45K_U$	$0.54K_U/T_U$	—
PID	$0.60K_U$	$1.2K_U/T_U$	$3K_UT_U/40$

Relay Auto Tuning



With more capable processors the auto tuning process could consist of:

1. Trying a small impulse signal and calculating tuning parameters from the response .
2. Looking at historical process data to derive the tuning parameters.
3. Starting with the Z-N values and trying small changes to look for improvements.
4. Basing the parameter on an entirely mathematical model of the process.